relsynth: Statistical Guarantees and Checks for Relational Synthetic Data

Mixed types, star and snowflake schemas, self-referencing hierarchies, temporal facts and text

Data Synthesisation project

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Builds on `algorithms.pdf` (layer 1). Code: `src/relsynth`; checks: `legacy/checks/run_checks.py` (C00–C78) and `tests/`

Abstract

Layer 1 fixed the methods and their complexity. This document states, for the most common synthesis situations, what the implemented model guarantees and proves it, then names the numerical check that verifies each statement on fake data with known ground truth (checks C00–C68; Table 1). Version 0.3 (Section 9) adds what the first real-database run required: grains (unique keys made of foreign keys), structural zeros, computed columns (lookups, derives, aggregates), nullable keys, a rank-likelihood copula and posterior predictive draws. The situations are: one table with mixed column types and missing values; star schemas; snowflake chains; self-referencing hierarchies; time-stamped facts; free text; business rules; and an enterprise schema combining all of them. Section 18 gives the checks that need no ground truth, which are the ones the local scripts run on real data, and Section 19 the diagnostics loop that turns their output into spec fixes. Sections 10–12 add what the next three real databases required. Section 13 makes every synthetic dataset a draw from the posterior predictive distribution and adds rules that hold on most rows, group effects, link affinity, participation hurdles, cannot-link strata, type 2 slowly changing dimensions, unknown members and bridges (checks C57–C68). Sections 14 and 15 add identification from the data, the exported model and a review of pooling and multiplicity (C69–C78). Section 16 (v1.0) fits fact tables larger than memory from statistics computed inside the database and proves that this leaves the relational model unchanged and the attribute models an exact posterior given a uniform sample.

Contents

1 Setting and notation **1**

2 One table with mixed types **2**

3 Star schemas **3**

4 Snowflake chains and joins **4**

5 Self-referencing hierarchies **5**

6 Temporal facts **5**

7 Text **6**

8 Business rules and the whole schema **6**

9 Grain, structural zeros and computed columns (v0.3) 7
9.1 Rules and computed columns . . . 7
9.2 Structural zeros . . . 8
9.3 Partial pooling . . . 9
9.4 Grain . . . 10
9.5 Dependence and posterior predictive draws . . . 11
9.6 Privacy on coarse tables . . . 12
9.7 Defaults . . . 12

10 Small operational databases (v0.4) 12
10.1 Rows per parent . . . 13
10.2 Hierarchies of categorical columns . . . 14
10.3 Tiny parents . . . 14
10.4 Stratification by larger code lists . . . 15
10.5 Values . . . 15
10.6 Introspection . . . 16
10.7 Defaults (v0.4) . . . 16

11 Data warehouses with order headers (v0.5) 16
11.1 Identifying the data model . . . 17
11.2 Generating dependent records . . . 18
11.3 Run time of the segment merge . . . 19
11.4 Structural zeros, mandatory participation and hierarchies . . . 19

12 Normalised and denormalised warehouses (v0.6) 21
12.1 Identifying the data model . . . 21
12.2 Columns that are functions of other columns . . . 23
12.3 Rules across tables . . . 24
12.4 Cost of stratification . . . 27

13 Bayesian synthesis and the whole data model (v0.7) 27
13.1 Posterior predictive synthesis . . . 27
13.2 Rules that hold on most rows . . . 28
13.3 Values shared by a group . . . 28
13.4 Two links that agree more often than chance . . . 28
13.5 Parents without children and segments that mix eligible rows . . . 29
13.6 Warehouse structures: slowly changing dimensions, unknown members, bridges . . . 30
13.7 Identifying the whole model . . . 31
13.8 The specification . . . 31

14 Identification from the data, exact speed-ups and the exported model (v0.8) 31
14.1 Keys and links from the data . . . 32
14.2 Exact speed-ups . . . 33
14.3 Run-length control, corrected . . . 33
14.4 The exported model . . . 34

15 Review: pooling, exact conditioning and calibrated limits (v0.9) 36
15.1 Structural blocks pooled across a relation . . . 36
15.2 Counts, inclusion and participation . . . 37
15.3 Exact conditioning on rules . . . 38
15.4 Copula sample size . . . 38
15.5 Identification . . . 39
15.6 Calibrated limits of the scorecard . . . 39

16 Fitting inside the database (v1.0) **40**
16.1 What the relational model needs from a fact . . . 40
16.2 Normal scores on database servers . . . 41
16.3 The sample . . . 42
16.4 Why entities are always loaded . . . 42
16.5 Sizes of child entities . . . 43

17 Evaluation thresholds **43**

18 Checks without ground truth (real data) **44**

19 Diagnostics loop **44**

20 Changes to layer 1 found while building and testing **45**

21 Limitations and open items **46**

22 Results **46**

1 Setting and notation

A database has entity tables e (rows $u \in U_e$, $n_e = |U_e|$) and fact tables f. Each entity row has a latent segment $z_u \in \{1..K_e\}$. Links are of five kinds (layer 1, Algorithm 3): *edges* (many-to-one between entities, e.g. product→category), *facts* with an anchor e_1 and a partner e_2, *extra* fact dimensions, *single-dimension* facts, and *trees* (self-references). Column values are mapped to a model space by a codec: discrete columns to codes $0..V-1$, continuous columns to reals (dates to days). Φ, ϕ are the standard normal cdf and density.

The implementation follows the layer 1 pseudocode, with the changes listed in Section 20. Every proposition below names the check that tests it.

2 One table with mixed types

Model. Within a segment (or the whole table), $\mathbf{z} \sim \mathcal{N}_p(0, C)$ and $x_j = F_j^{-1}(\Phi(z_j))$ with $F_j^{-1}(u) = \inf\{x : F_j(x) \geq u\}$. A missing value is the lowest "level" of the latent scale: with null rate q_j,

$$x_j = \begin{cases} \text{null}, & \Phi(z_j) < q_j, \\ G_j^{-1}((\Phi(z_j) - q_j)/(1 - q_j)), & \text{otherwise}, \end{cases} \tag{1}$$

where G_j is the distribution of the non-null values.

Proposition 1 (Marginal preservation)**.** *If C has unit diagonal, then for every j the synthetic x_j has exactly the distribution $q_j\delta_{null} + (1 - q_j)G_j$, whatever C is. In particular the null rate is q_j and discrete columns have their level probabilities exactly.*

Proof. $z_j \sim \mathcal{N}(0, 1)$, so $U = \Phi(z_j)$ is uniform on $(0, 1)$ (probability integral transform). $P(\text{null}) = P(U < q_j) = q_j$. Given $U \geq q_j$, $V = (U - q_j)/(1 - q_j)$ is uniform, and $P(G_j^{-1}(V) \leq x) = P(V \leq G_j(x)) = G_j(x)$ by the Galois inequality $G^{-1}(v) \leq x \iff v \leq G(x)$, which holds for any cdf, including step functions. □

Check C01: KS or total-variation distance between real and synthetic, per column, is within the two-sample critical value at the 1% level (Section 17).

Bayesian marginals. The synthesiser draws F_j from its posterior, so that synthetic data reflect estimation uncertainty (posterior predictive synthesis):

- discrete: $\mathbf{p} \sim \mathrm{Dir}(\mathbf{c} + a\,\mathbf{p}_0)$ with exact level counts $\mathbf{c}$ and $\mathbf{p}_0$ the parent marginal (the whole table for a segment);
- continuous: Dirichlet-process posterior $F \sim \mathrm{DP}\left(a+n,\ (aF_0 + \sum_i \delta_{x_i})/(a+n)\right)$, drawn as $F = V \cdot \sum_i w_i \delta_{x_i} + (1-V)F_0$ with $w \sim \mathrm{Dir}(1,\ldots,1)$ (Bayesian bootstrap, Rubin 1981) and $V \sim \mathrm{Beta}(n, a)$;
- null rate: $q \sim \mathrm{Beta}(n_{\text{null}} + aq_0,\ n - n_{\text{null}} + a(1-q_0))$.

Proposition 2 (Posterior mean of a segment marginal)**.** *Under the draws above,* $\mathbb{E}[F(t) \mid \textit{data}] = (aF_0(t) + nF_n(t))/(a+n)$ *for every* t*, where* F_n *is the empirical cdf of the segment. Small segments are therefore shrunk towards the parent marginal, large ones follow their own data.*

Proof. $\mathbb{E}[V] = n/(n+a)$, $\mathbb{E}[\sum_i w_i \mathbf{1}\{x_i \le t\}] = F_n(t)$ because $\mathbb{E}w_i = 1/n$, and V is independent of w. The draw is the stick-breaking-free representation of the DP posterior (Ferguson 1973): the posterior base measure is the stated mixture, and the $\mathrm{DP}(a+n, \cdot)$ restricted to the atoms gives Dirichlet weights proportional to $(1,\ldots,1,a)$, which is what $(Vw, 1-V)$ is. □

Check C17 (400 draws, closed form within 0.03). The prior strength a is `marginals.dirichlet_prior` (default 0.25: a segment of 6 rows keeps 96% of its own distribution).

Correlation. Stage 2 of layer 1's Algorithm 2: latent bins $(\ell_{ij}, u_{ij}]$ from the full-data marginals, data augmentation $z_{ij} \mid \cdot \sim \mathrm{TN}(\mu_{ij}, \sigma_j^2; \ell_{ij}, u_{ij})$, and Metropolis on the C-vine partial correlations with the LKJ(η) prior.

Proposition 3 (The sampler targets the copula posterior)**.** *The chain alternating (i) the column-wise truncated-normal updates of* Z *and (ii) Metropolis–Hastings steps on* $y = \mathrm{atanh}(w)$ *with target* $\pi(y) = \sum_k (b_k - 1)\log(1-w_k^2) - \frac{n}{2}\log|C| - \frac{1}{2}\mathrm{tr}(C^{-1}Z^\top Z) + \sum_k \log(1-w_k^2)$ *(up to constants,* b_k *the LKJ Beta shapes) has the joint posterior* $p(C, Z \mid \textit{bins})$ *as its invariant law.*

Proof. (i) is an exact Gibbs step: given C and the other columns, z_{ij} is normal with mean $-\sum_{k\neq j} \Omega_{jk} z_{ik}/\Omega_{jj}$ and variance $1/\Omega_{jj}$ ($\Omega = C^{-1}$), restricted to its bin. (ii) The C-vine map $w \mapsto C$ is a bijection from $(-1,1)^{p(p-1)/2}$ onto correlation matrices, and under LKJ(η) the w_k are independent with $(w_k+1)/2 \sim \mathrm{Beta}(b_k, b_k)$ (Lewandowski, Kurowicka and Joe 2009); $\log(1-w_k^2)$ in π combines the Beta density and the Jacobian $dw/dy = 1 - w^2$. Random-walk MH with a symmetric proposal leaves π invariant; the adaptive proposal covariance is frozen after the burn-in, so the post-burn-in chain is an ordinary MH chain (Haario, Saksman and Tamminen 2001). A Gibbs/MH cycle of kernels each invariant for the joint law is invariant for it. □

Check C02: on 4,000 rows from a known 7-dimensional copula with a binary, a 4-level nominal, an ordinal, a date and a 12%-missing column, 20 of 21 true correlations fall in the 99% posterior intervals; posterior means are within 0.08; split $\hat{R} \le 1.1$.

Remark 1 (Dependent missingness). Because the null is the lowest latent bin of its column, $P(\text{null}_j \mid x_k)$ inherits the dependence of z_j on z_k: missingness that is monotone in another column is reproduced (Check C03: P(cholesterol missing | age quartile) within 0.025). Non-monotone missingness needs a separate missingness indicator column, which the spec can add.

Remark 2 (Nominal columns). A Gaussian copula needs an order on nominal levels. Two devices keep dependence: (a) levels are ordered by the mean normal score of the most associated other column (largest correlation ratio), which makes the main dependence monotone; (b) two low-cardinality categoricals with Cramér's $V \ge 0.3$ and at most 64 joint levels are modelled as one categorical over their combinations, i.e. a Dirichlet–categorical clique as in a Bayesian network (layer 1 item 4). By Proposition 1 their joint law is then reproduced exactly.

3 Star schemas

For a fact f between anchor e_1 and partner e_2, the count of rows for the pair (u, v) is $c(u, v) \sim \text{Pois}(\eta_{z_u z_v})$ independently.

Proposition 4 (Identifiability of segments)**.** *Let m_l be the number of partner rows in segment l, and suppose the vectors $r_k = (\eta_{kl} m_l)_l$ are pairwise distinct. Then, given the partner partition, the anchor partition is identified up to relabelling from the expected row-statistic profiles $\mathbb{E}[C^{\text{row}}_{u\cdot}] = r_{z_u}$, and the conditional of z_u concentrates on the true segment as the row's total count grows.*

Proof. $C^{\text{row}}_{ul} = \sum_{v: z_v = l} c(u, v) \sim \text{Pois}(\eta_{z_u l} m_l)$, independent over l. Distinct r_k give distinct laws of $C^{\text{row}}_{u\cdot}$, so the map $k \mapsto$ law is injective and the partition is determined up to a permutation of labels. For the posterior: by layer 1 Proposition 1, $\log p(z_u = k) - \log p(z_u = k') = \sum_l [C^{\text{row}}_{ul} \log(\eta_{kl}/\eta_{k'l}) - (\eta_{kl} - \eta_{k'l}) m_l] + O(1)$. Its expectation under $z_u = k$ equals $\sum_l m_l \,\text{KL}(\text{Pois}(\eta_{kl}) \| \,\text{Pois}(\eta_{k'l})) > 0$ times the exposure, and it grows linearly in m_l while its standard deviation grows as $\sqrt{m_l}$. □

Check C05: with caps of 8, the sampler and merge recover 4 customer segments (ARI 0.98) and 3 product segments (ARI 1.0).

Proposition 5 (Block-rate posterior)**.** *Given the partitions, $\eta_{kl} \mid \cdot \sim \text{Gamma}(a + C^{\text{blk}}_{kl},\ b + n_k m_l)$, with mean $\to C^{\text{blk}}_{kl}/(n_k m_l)$ and coefficient of variation $(a + C^{\text{blk}}_{kl})^{-1/2}$.*

Proof. Gamma–Poisson conjugacy over the $n_k m_l$ independent pair counts of the block, whose sum is C^{blk}_{kl}; zero pairs enter through the exposure only. □

Check C06: all 12 true block rates lie in their 99% posterior intervals.

Proposition 6 (Splitting and degree correction)**.** *(a) Drawing $N \sim \text{Pois}(\eta W_1 W_2)$ rows for a block and assigning each to (u, v) with probabilities $\omega_u / W_1 \cdot \omega'_v / W_2$ ($W_1 = \sum \omega_u$, $W_2 = \sum \omega'_v$) gives independent $c(u, v) \sim \text{Pois}(\eta\, \omega_u \omega'_v)$. (b) If $\omega_u \sim \text{Gamma}(\alpha, \alpha)$, the number of rows of an anchor row with expected count μ is negative binomial with $\mathbb{E} = \mu$ and $\text{Var} = \mu + \mu^2/\alpha$. (c) Rescaling the ω of a segment to sum to the segment size leaves every block's expected total unchanged, so the scale rule (Proposition 7) stays exact.*

Proof. (a) Poisson colouring: a Poisson total thinned multinomially gives independent Poisson cells with means $\eta W_1 W_2 \cdot \omega_u \omega'_v / (W_1 W_2)$. (b) $\mathbb{E}[N \mid \omega] = \mu\omega$, so $\mathbb{E}N = \mu$ and $\text{Var}\, N = \mathbb{E}\,\text{Var}(N \mid \omega) + \text{Var}\,\mathbb{E}(N \mid \omega) = \mu + \mu^2/\alpha$. (c) The expected total is $\eta W_1 W_2$, and after rescaling $W_1 = n_k$, $W_2 = m_l$ as without frailty. □

The frailty (a degree-corrected block model, Karrer and Newman 2011) is switched on per segment when the dispersion index of per-row totals exceeds 1.5 and the segment has at least 30 rows. *Checks C07* (rows-per-customer distribution within the KS critical value; repeated-pair share equal) *and C16* (variance formula within 2%).

Proposition 7 (Scale rule)**.** *With $\eta^*_{kl} = \eta_{kl} m_l / m^*_l$, the expected number of rows per anchor row is unchanged when any table is resized; resizing the anchor by s multiplies the fact's expected size by s.*

Proof. Layer 1, Proposition 4: $\sum_l \eta^*_{kl} m^*_l = \sum_l \eta_{kl} m_l$. With $n^*_k = s n_k$, $\mathbb{E}N^* = \sum_{kl} \eta^*_{kl} n^*_k m^*_l = s \sum_{kl} \eta_{kl} n_k m_l$. □

Check C08: products ×2 and ×0.5 keep rows per customer within 4%; customers ×2 doubles the fact (2.03).

4 Snowflake chains and joins

An edge child→parent is the same Poisson block model with $c(u, v) \in \{0, 1\}$. At synthesis each child row picks its parent segment from $\varphi_{kt} \propto \eta_{kt} m_t$, then a parent inside the segment (with frailty weights).

Proposition 8 (Chained segments compose)**.** *For a chain fact→product→category→department, the synthetic law of the department segment of a fact row with product segment l is $\sum_t \varphi_{lt}^{(p\to c)} \varphi_{ts}^{(c\to d)}$. If the source satisfies the Markov property "a row's ancestor two levels up depends on the row only through its parent's segment", then every joint law of a fact attribute and an ancestor attribute is reproduced in expectation.*

Proof. Each synthetic step draws the next segment from φ given the current segment only, so the chain of segments is a Markov chain whose transition matrices are the fitted φ; the two-step law is their product (Chapman–Kolmogorov). Attributes are drawn given their own row's segment. Under the stated Markov property the source has the same factorisation, so the joint law of (fact attribute, ancestor attribute) is $\sum$ over segment paths of the product of the same factors. □

Fact measures that depend on a parent's value. Segments carry a parent's attributes only coarsely; a sale amount that follows the product's own price needs more. For every fact, continuous attributes of its parents (at most two per parent) are joined onto the fact rows and enter the fact's copula as conditioning columns a. At synthesis the parent is chosen first; its synthetic attribute values give z_a, and the fact's own columns are drawn from $z_r \mid z_a \sim \mathcal{N}(C_{ra}C_{aa}^{-1}z_a,\ C_{rr} - C_{ra}C_{aa}^{-1}C_{ar})$, which is the exact Gaussian conditional, so Proposition 1 holds for the pair (fact column, parent column) jointly.
Check C09: segments of department, category and customer are recovered (ARI 1.0, 1.0, 0.997). Three-hop statistics (share of sales by department division, mean amount per division, Spearman correlation of amount with the product's price) fall inside the posterior predictive spread of 20 synthetic databases ($|z| \le 3$; the amount–price correlation is 0.909 real vs 0.910 synthetic).

5 Self-referencing hierarchies

A self-reference (employee→manager) is a multi-type Galton–Watson process: a node of type k has $N \sim \text{Pois}(\lambda_k)$ children (with optional Gamma frailty), each of type l with probability ψ_{kl}; roots have types $\text{Cat}(\pi^{\text{root}})$. Let $M_{kl} = \lambda_k \psi_{kl}$ be the mean matrix.

Proposition 9 (Generation sizes, finiteness and expected size)**.** *Let r be the row vector of expected root counts per type and Z_n the vector of type counts in generation n. Then $\mathbb{E}Z_n = rM^n$. If the spectral radius $\rho(M) < 1$, every tree is finite almost surely and $\mathbb{E}|T| = r(I - M)^{-1}\mathbf{1}$.*

Proof. $\mathbb{E}[Z_{n+1} \mid Z_n] = Z_n M$ by linearity over parents, so $\mathbb{E}Z_n = rM^n$ by induction. If $\rho(M) < 1$ then $\sum_n M^n = (I - M)^{-1}$ converges, so $\mathbb{E}\sum_n Z_n\mathbf{1} = r(I - M)^{-1}\mathbf{1} < \infty$, and a non-negative variable with finite mean is finite a.s. □

Proposition 10 (Fit and validity)**.** *(a) Given the types, the posteriors are $\psi_k \sim \text{Dir}(\frac{1}{2} + B_{k\cdot})$ with B_{kl} the number of type-k parents' type-l children, and $\lambda_k \sim \text{Gamma}(a + \sum_{u:z_u=k} N_u,\ b + n_k)$. (b) Nodes at even depth are conditionally independent given nodes at odd depth, so the two-colour blocked Gibbs sweep is valid. (c) Synthesis generates each node once, as the child of an already generated node, so the synthetic hierarchy is a forest: every non-root has exactly one existing parent and there are no cycles. Growth stops at a maximum depth (by default the deepest level of the source, so no node is deeper than any source node;* `trees.max_depth` *overrides it).*

Proof. (a) Multinomial–Dirichlet and Poisson–Gamma conjugacy. (b) The type of u appears only in factors with its parent and its children, whose depths have the other parity. (c) By construction of the breadth-first generation. □

Checks C10, C11: types recovered (ARI 1.0); $\rho(\hat{M})$ within 0.05 of the source tree's realised value and below 1; the formula for $\mathbb{E}|T|$ agrees with 300 simulations within three standard errors; synthetic hierarchies have no cycles and no dangling managers; the depth histogram passes a posterior predictive check against 40 re-simulations ($p \geq 0.05$); children-per-node KS 0.016; grade given depth within 0.10.

Remark 3. Near-critical hierarchies ($\rho(M)$ close to 1, as in the test tree: managers managing managers with mean 0.9) have depth profiles that vary a lot between realisations. A fixed tolerance on the depth histogram is then wrong; the posterior predictive check (Gelman, Meng and Stern 1996) is the right test, and is what the evaluator uses.

6 Temporal facts

Rows carry a time bucket b (day/week/month/quarter/year). Following layer 1's design, $c(u, v, b) \sim \text{Pois}(\eta_{kl}\lambda_{kb}\Delta_b)$ with $\sum_b \lambda_{kb}\Delta_b = \sum_b \Delta_b$.

Proposition 11 (Seasonality by multinomial thinning)**.** *Given a block total N_{kl}, bucket counts are* $\text{Mult}(N_{kl};\ \theta_{k\cdot})$ *with $\theta_{kb} \propto \lambda_{kb}\Delta_b$; the posterior of the profile is $\theta_k \sim \text{Dir}(\theta_0 + n_{k\cdot})$ with n_{kb} the source rows of anchor segment k in bucket b. Synthesis that draws a bucket per row from θ_k, then a uniform instant inside the bucket, therefore has the time law of the model.*

Proof. Independent Poisson cells conditioned on their sum are multinomial with probabilities proportional to their means; the factor η_{kl} cancels. Dirichlet–multinomial conjugacy gives the posterior. □

Check C12: two customer segments with opposite seasonality are recovered (ARI 0.995); fitted monthly profiles are within TVD 0.05 of the true ones; the synthetic monthly histogram is within 0.03 overall and 0.04 within each plan.

7 Text

Free-text columns use a class-conditional hierarchical Dirichlet back-off n-gram model (MacKay and Peto 1995). With h_n the last n tokens, c the value of the conditioning column (or the row's segment) and p_G the add-one global unigram,

$$p_c(w \mid h_n) = \frac{n_c(h_n, w) + \beta\, p_c(w \mid h_{n-1})}{n_c(h_n) + \beta}, \qquad p_c(w \mid h_0) = \frac{n_c(w) + \beta\, p_G(w)}{n_c + \beta}. \tag{2}$$

Proposition 12 (Proper predictive)**.** *For every class and context, $p_c(\cdot \mid h_n)$ is a probability distribution on the vocabulary, and it is the posterior predictive of a Dirichlet prior with mean $p_c(\cdot \mid h_{n-1})$ and concentration β on the next-token distribution after h_n.*

Proof. Induction on n: p_G sums to one; if $p_c(\cdot \mid h_{n-1})$ does, the numerator of (2) sums to $n_c(h_n) + \beta$. The second statement is Dirichlet–multinomial conjugacy. Contexts never seen use the shorter context, which is the same formula with zero counts. □

Proposition 13 (Memorisation guards)**.** *Let k be* `min_token_rows` *and L* `max_copy_ngram`*. (a) Every token in synthetic text occurs in at least k distinct source rows. (b) No synthetic text contains a run of L tokens that occurs in fewer than k source rows, nor equals a source text shared by fewer than k rows. (c) Accepted texts are distributed as the model law conditioned on the guard event.*

Proof. (a) Rarer tokens are removed from the vocabulary before counting, so they have probability zero. (b) Candidates that violate the rule are rejected. (c) Rejection sampling: a candidate is accepted with probability $\mathbf{1}\{\text{guard}\}$, so accepted draws have density proportional to $p \cdot \mathbf{1}\{\text{guard}\}$. After `max_retries` failures the cell is left empty rather than emitting a violating text. □

Emails and long digit runs are replaced by typed placeholders before training and refilled with fresh values. These guards bound verbatim memorisation; they are not differential privacy. For DP text the layer 1 route (Private Evolution over a foundation model, Xie et al. 2024) stays the option, and the same guards apply to its output. *Check C13*: the predictive sums to one within 10^{-9} over 125 random contexts; none of the uniquely named tickets' names appears; no rare token or rare 8-gram is copied; a classifier trained on synthetic tickets predicts the real ticket category as well as one trained on real tickets (ratio 1.0).

8 Business rules and the whole schema

Rules come from the spec: `derive` rules (`total = quantity * unit_price - discount`) are computed after sampling; `check` rules are enforced by redrawing the attributes of violating rows from the same model (same segment and parents).

Proposition 14 (Rules by rejection)**.** *If a row's attributes are drawn from p and redrawn until the check A holds, the result has law $p(\cdot \mid A)$, and the expected number of draws is $1/p(A)$. Derived columns satisfy their equation exactly.*

Proof. Standard rejection sampling with acceptance indicator $\mathbf{1}_A$; the number of draws is geometric with success probability $p(A)$. □

Checks C14, C15: on the 9-table enterprise schema (regions, customers, stores, a three-level product hierarchy, a 4-dimension sales fact with dates, tickets with text, and an employee hierarchy) the full scorecard passes, all 9 foreign keys resolve, derived totals are exact and no check rule is violated; on a test distribution the redrawn sample matches the exact truncated law (KS 0.003). *Check C18*: the same seed gives a bit-identical database.

Reference tables. Small lookup tables (regions, departments) are usually not personal data and their individual rows matter (one region is far bigger than another). A table with `mode: copy` is kept as is; each of its rows is its own fixed segment, so Proposition 5 then models every child's choice of that exact parent, which reproduces each parent's popularity. Introspection proposes this for tables of at most 100 rows whose parents are also copied.

9 Grain, structural zeros and computed columns (v0.3)

The first real-database run (a business model of a supervisory reporting framework: reporters, submissions, templates, cells and reported values) failed in ways the fake scenarios had not exercised: duplicate reports per reporter and date, duplicate values on fixed cells, templates filed partially or twice, values in cells of the wrong framework version, and a stored count that no longer matched the rows it counts. None of these is specific to that model. Each comes from one of four general situations: a table has a *grain* (a unique key made of foreign keys); a combination of parents is *structurally absent* (never allowed) rather than rare; a column is *computed* from other tables; or a key is *partly null*. This section gives the model for each and proves what it guarantees. Scenario S8 (a generic reporting model with known truth: reporters, periods, two framework versions, forms with sheets, items with open rows, omitted and partial filings, an aggregate and a derived status) tests them together.

9.1 Rules and computed columns

Rules are parsed into a restricted syntax tree (no attribute access, no calls outside a fixed list), evaluated column-wise with null semantics. The string functions (`regex_replace`, `regex_match`, `substr`, `concat`, `upper`, `lower`, `strip`, `length`) require constant patterns.

Proposition 15 (Three-valued rules agree with the database)**.** *Logical connectives are evaluated in Kleene's strong three-valued logic, comparisons with a null are unknown,* `where` *takes its else branch on unknown, functions of a null are null, and a* `check` *fails only when it evaluates to false. These are the semantics of SQL* `AND`*,* `OR`*,* `NOT`*,* `CASE` *and* `CHECK`*; hence a row accepted by a database* `CHECK` *constraint is accepted by the same rule here, and conversely.*

Proof. SQL defines `AND`/`OR`/`NOT` by the strong Kleene truth tables (ISO/IEC 9075-2, 8.20) and a `CHECK` constraint as violated only when its condition is false. The evaluator implements exactly these tables elementwise (unknown is a third value, not false), so both evaluations agree row by row. □

Check C20: the three connectives on all nine pairs of truth values, nulls in `where`, `between`, `concat` and `regex_replace`, and refusal of non-constant patterns and of calls outside the list.

A column is *computed* when it is a `lookup` (`fk.col`, possibly several hops), a `derive` rule, or an `aggregate` rule (count, sum, min, max, mean, exists or count of distinct values of the child rows that reference the row, through a plain or a composite key). Computed columns form a directed dependency graph G: $c \to c'$ when the rule of c' reads c.

Proposition 16 (Computed columns are exact)**.** *(a) If G is acyclic, the fixed-point pass of* `prepare_source` *fills every computed column missing from the source, each with the value of its rule on the final inputs, and reports, for each stored computed column, the share of rows where the stored value differs from its rule. (b) On the synthetic database every computed column is recomputed after the tables it reads are generated (aggregates and the derives that read them last, then lookups and derives of the dependent tables again), so every derive, lookup and aggregate identity holds exactly. If G has a cycle, the columns on it are reported and no value is invented.*

Proof. (a) Each pass visits tables in foreign-key order and computes a column once all columns it reads exist; a column at depth d of G is therefore filled by pass $d+1$, and a pass without progress certifies that the remaining columns lie on or above a cycle. Values are computed from the final inputs because inputs are filled before readers. (b) Generation never samples a computed column; the late recomputation runs in topological order of G restricted to synthetic tables, so each value equals its rule applied to the final values it reads. □

Checks C26, C32: on S8 the missing `form_code` is filled from its rule, 10% planted mismatches in a stored count are reported as 0.0976 (the planted share), and on the synthetic data the count, the derived status and the form code match their rules on every row.

9.2 Structural zeros

A real schema has combinations that never occur because they are not allowed (a cell of another framework version, a header item missing from a filed form, a null in a mandatory column), mixed with combinations that are merely rare. A Beta or Dirichlet prior gives every combination positive probability, so with enough synthetic rows the forbidden ones appear. A spike-and-slab prior separates the two.

Proposition 17 (Spike-and-slab Bernoulli)**.** *Let* $p \sim \frac{\pi_s}{2}\delta_0 + \frac{\pi_s}{2}\delta_1 + (1-\pi_s)\,\mathrm{Beta}(a,b)$ *and* $S \mid p \sim \mathrm{Bin}(T,p)$*. The posterior is*

$$p \mid S \;\propto\; \tfrac{\pi_s}{2}[S=0]\,\delta_0 + \tfrac{\pi_s}{2}[S=T]\,\delta_1 + (1-\pi_s)\tfrac{B(a+S,b+T-S)}{B(a,b)}\,\mathrm{Beta}(a+S,\,b+T-S),$$

with mean $w_1 + w_2\,\frac{a+S}{a+b+T}$ *for the normalised weights* w_0, w_1, w_2*. The same holds for a null rate (*`_null_rate`*) and for each occupancy probability (*`zoib_draw`*).*

Proof. Bayes' rule with a mixture prior: each component's posterior weight is its prior weight times its marginal likelihood; the binomial coefficient is common to all three and cancels; the point masses have likelihood $[S=0]$ and $[S=T]$, the slab $B(a+S,b+T-S)/B(a,b)$. □

When the rows k of a group g (the segments of one stratum, e.g. the submissions of one framework version) share the structure, the evidence must be pooled: each row alone may have seen a combination only a few times.

Proposition 18 (Grouped structural support)**.** *Let $G_{gl} \in \{\text{all zero}, \text{all one}, \text{free}\}$ with prior $(\frac{\pi_s}{2}, \frac{\pi_s}{2}, 1-\pi_s)$, $p_{kl} = 0$ (resp. 1) for all $k \in g$ if G_{gl} is all zero (all one), and p_{kl} independent with the prior of Proposition 17 if free. Then $P(G_{gl} \mid S) \propto (\frac{\pi_s}{2}\prod_k [S_{kl} = 0], \ \frac{\pi_s}{2}\prod_k [S_{kl} = T_{kl}], \ (1-\pi_s)\prod_k m_{kl})$ with m_{kl} the per-row marginal of Proposition 17, and given "free" the p_{kl} follow their per-row posteriors.* `grouped_zoib_draw` *samples this posterior exactly. In particular, a combination never observed in a group is absent from every row of the group with posterior probability tending to 1 as the number of rows with exposure ($T_{kl} \geq 1$) grows, even when every T_{kl} stays bounded.*

Proof. The likelihood factorises over k given G_{gl}; under the two pooled states it is the product of indicators, under "free" the product of per-row marginals because the p_{kl} are independent. Given "free", the posterior of $(p_{kl})_k$ factorises, which is what the per-row draw returns. For the limit: if all $S_{kl} = 0$ and $T_{kl} \geq 1$, then $m_{kl} = \frac{\pi_s}{2} + (1-\pi_s)B(a, b+T_{kl})/B(a,b) \leq \frac{\pi_s}{2} + (1-\pi_s)\frac{b}{a+b} < 1$, because $B(a, b+T)/B(a,b)$ decreases in T and equals $b/(a+b)$ at $T = 1$. So $\prod_k m_{kl} \to 0$ geometrically while the "all zero" weight $\pi_s/2$ is constant. A group without exposure gets its prior back ($m_{kl} = 1$, both indicators hold). □

This is used for the occupancy probabilities p_{kl} and for the anchor activity π (which parents have any rows) with the strata as groups.

Proposition 19 (Sparse Dirichlet, exact inclusion)**.** *Let each level j be structurally absent with prior probability π_s independently, $p_J \sim \mathrm{Dir}(\alpha_J)$ on the included set J, and counts $c \sim \mathrm{Mult}(n, p)$. Observed levels are included; for the set Z of unobserved levels, with $\rho = (1-\pi_s)/\pi_s$, $A_o = \sum_{c_j>0} \alpha_j$,*

$$P(J \cap Z = I \mid c) \ \propto \ \rho^{|I|} \, \frac{\Gamma(A_o + \alpha_I)}{\Gamma(A_o + \alpha_I + n)}, \qquad \alpha_I = \textstyle\sum_{j \in I} \alpha_j,$$

and with an auxiliary $u \in (0,1)$ of density $\propto u^{A_o - 1}(1-u)^{n-1}\prod_{j \in Z}(1 + \rho u^{\alpha_j})$, the indicators $[j \in J]$ are independent given u with probabilities $\rho u^{\alpha_j}/(1+\rho u^{\alpha_j})$. `sparse_dirichlet_draw` *samples u by inverse cdf on a grid of* $\mathrm{logit}\, u$*, then J, then $p_J \sim \mathrm{Dir}(c_J + \alpha_J)$.*

Proof. The Dirichlet-multinomial likelihood is $\frac{n!}{\prod c_j!}\frac{\Gamma(A_J)}{\Gamma(A_J+n)}\prod_{j \in J}\frac{\Gamma(\alpha_j + c_j)}{\Gamma(\alpha_j)}$; unobserved levels contribute factor 1 to the product, giving the display. Since $\Gamma(x)/\Gamma(x+n) = B(x,n)/\Gamma(n) = \Gamma(n)^{-1}\int_0^1 u^{x-1}(1-u)^{n-1}du$, the joint of (u, I) is $\propto u^{A_o - 1}(1-u)^{n-1}\prod_{j \in I}\rho u^{\alpha_j}$, whose I-marginal is the display and whose conditional factorises over j. Given J, conjugacy gives the Dirichlet. □

The earlier one-level-at-a-time Bayes factor is exact only when one level is unobserved; with five unobserved levels it under-stated absence by up to 0.07 (rejection reference), which is why it was replaced. *Check C21*: zoib, grouped zoib, sparse Dirichlet (one and five unobserved levels) and null rates agree with exact posteriors obtained by rejection sampling from the generative model (all within Monte Carlo error, e.g. 0.857 vs 0.857, 3.242 vs 3.228 absent levels).

9.3 Partial pooling

Proposition 20 (Empirical-Bayes Beta slab)**.** *For row k of a block matrix of presence counts, on its supported entries ($S_{kl} > 0$) let $m = \sum S / \sum T$ and $\hat\tau^2 = \widehat{\mathrm{Var}}(S_l/T_l) - \overline{m(1-m)/T_l}$. If $p_{kl} \sim \mathrm{Beta}(cm, c(1-m))$ with $S_{kl} \sim \mathrm{Bin}(T_{kl}, p_{kl})$, then $\hat c = m(1-m)/\hat\tau^2 - 1$ is the moment estimator of c (Kleinman 1973): as the number of entries grows it converges to c up to a relative error $O(1/\bar T)$. The slab $\mathrm{Beta}(\hat c m, \hat c(1-m))$, bounded to $\hat c \in [1, 10^4]$, shrinks entries whose rates differ only by binomial noise towards m and leaves real differences.*

Proof. $\mathbb{E}[S/T] = m$ and $\mathrm{Var}(S/T) = \mathbb{E}\,\mathrm{Var}(S/T \mid p) + \mathrm{Var}(p) = \mathbb{E}[p(1-p)]/T + \mathrm{Var}(p)$ with $\mathrm{Var}(p) = m(1-m)/(c+1)$ and $\mathbb{E}p(1-p) = m(1-m) - \mathrm{Var}\,p$. Hence $\mathrm{Var}(S/T) - m(1-m)/T = \mathrm{Var}(p)\,(1 - 1/T)$, so $\hat{\tau}^2 \to \mathrm{Var}(p)(1 + O(1/\bar{T}))$ by the law of large numbers, and $c = m(1-m)/\mathrm{Var}(p) - 1$ gives the limit of $\hat{c}$ by continuity. The bounds keep the prior proper and stop $\hat{\tau}^2 \le 0$ (no excess variance) from producing an infinite c. □

Proposition 21 (Multiplicities: spike-and-slab Poisson)**.** *Excess rows X_g over E_g present pairs of a group g (partner, anchor or block) have prior $\lambda_g \sim \pi_s\delta_0 + (1-\pi_s)\,\mathrm{Gamma}(a_m, b_m)$ and $X_g \mid \lambda_g \sim \mathrm{Pois}(\lambda_g E_g)$. The collapsed marginal is $\pi_s[X_g = 0] + (1-\pi_s)\frac{b_m^{a_m}\Gamma(a_m+X_g)}{\Gamma(a_m)(b_m+E_g)^{a_m+X_g}}$ per group (up to a factor common to all groupings), the grouping with the highest product of these over its groups is chosen, and b_m is set so that the slab mean a_m/b_m equals $\sum_{X_g>0} X_g / \sum_{X_g>0} E_g$: the groups with $X_g > 0$ are certainly in the slab, so pooling the spikes into the mean would dilute it.*

Proof. Each present pair i of group g has $x_i \sim \mathrm{Pois}(\lambda_g)$, so the likelihood is $\lambda_g^{X_g} e^{-\lambda_g E_g} / \prod_i x_i!$; Poisson-Gamma conjugacy gives the slab marginal shown and the mixture follows as in Proposition 17. The factor $\prod_i 1/x_i!$ is the same for every grouping of the same pairs, so comparing groupings is a Bayes-factor comparison. Groups with $X_g > 0$ have zero spike likelihood, so they belong to the slab; selecting on $X_g > 0$ raises the estimate by at most the factor $1/(1-(b_m/(b_m+E_g))^{a_m})$, which is close to 1 for the exposures used. □

Checks C22, C28: the concentration is recovered (median 20.5 for true 20), shrinkage lowers the error of p by 27%, homogeneous rates pool almost fully; S8 open rows repeat 4.10 times on average (truth 4) and the partner grouping is selected.

9.4 Grain

Proposition 22 (Repeat test)**.** *Under the degree-corrected Poisson model with $\lambda_{ij} = r_i s_j / N$ (plug-in row and column totals), the number of pairs with two or more rows is a sum of independent Bernoulli variables with probabilities $q_{ij} = 1 - e^{-\lambda_{ij}}(1+\lambda_{ij})$, so $z = (obs - \sum q)/\sqrt{\sum q(1-q)}$ is asymptotically standard normal (Lyapunov). Under an occupancy model with a unique key, obs $= 0$. The engine selects occupancy when $z < -4$ and $\sum q \ge 10$, or when a unique key on the pair is declared.*

Proof. Counts of distinct pairs are independent Poisson variables; $P(\text{count} \ge 2) = q_{ij}$; the variances are bounded and their sum diverges with the number of pairs, which is Lyapunov's condition. □

Remark 4. A block structure on top of the degree correction concentrates the rates, and $q(\lambda) \approx \lambda^2/2$ is convex, so it tends to raise the expected repeats: the rank-one reference is the conservative choice. On S8 the filing and value tables give $z = -29$ and $z = -61$; the S2 sales fact keeps the Poisson model (check C24).

Proposition 23 (Grain entities)**.** *Let an entity table R have a unique key (a, b) of two foreign keys to A and B. The model: pair (a, b) is present with probability $p_{z_a z_b}$ (occupancy, priors of Propositions 18 and 20), and the row's own segment is drawn from $P(z \mid z_a, z_b) \propto \pi_z\, \phi^A_{z z_a} \phi^B_{z z_b}$. Then (a) R^* has no duplicate key; (b) the expected number of rows is $\sum_{a,b} p_{z_a z_b}$, which scales with $s_A s_B$ when A, B are scaled by s_A, s_B (a copied B has $s_B = 1$); (c) the segment draw is the exact conditional of a latent class model in which z_a and z_b are independent given z.*

Proof. (a) Each pair is generated at most once. (b) Linearity of expectation over the $n_A n_B$ pairs, whose segment frequencies are preserved by scaling (Proposition 7). (c) Bayes' rule with prior π and the two conditionally independent likelihood factors. □

Checks C26, C29: no duplicate report at scale 1 or 2; reports per reporter 7.675 (real 7.650) at scale 2 with the periods copied.

Proposition 24 (Unique keys: de-duplication is exact conditioning)**.** *Let a pair be present with probability p and, when present, carry $M = 1 + N$ rows with exchangeable attributes. Keeping the first row of each pair gives (a) presence exactly* Bern(p)*, (b) kept attributes with exactly the law of one row. For a filtered key (`unique: {columns, where}`) the same holds on the rows that satisfy the filter, which the occupancy model generates with multiplicity one; rows outside the filter (open rows) keep their multiplicity.*

Proof. (a) $\{M \geq 1\} = \{\text{present}\}$. (b) By exchangeability the first row has the marginal law of any row and does not depend on N. □

Checks C23, C28: presence 0.3522 for $p = 0.35$ (within 3 s.d.) and KS 0.005 for the kept rows; on S8 no closed item is ever duplicated while open rows keep their multiplicity.

Proposition 25 (Strata)**.** *Let every segment of an entity belong to exactly one level of a stratum key (e.g. the framework version of a submission). (a) Setting the key from the segment makes every synthetic (segment, level) pair one that occurs in the real data. (b) With the strata as groups in Proposition 18, a partner never seen with a level is absent from that level with posterior probability tending to 1 as the level's exposed segments grow; for grain entities the segment draw is also restricted to segments compatible with both parents' levels, which is exact.*

Proof. (a) The key is a function of the segment. (b) Proposition 18; the restriction of the segment draw puts zero probability outside the compatible set. □

Check C26: the item→version link of S8 is never declared, yet every synthetic value uses an item of its report's version (share 1.0).

Proposition 26 (Nullable extra keys)**.** *Let a fact row have anchor and partner keys and further keys that may be null. If rows with a null extra key are dropped before fitting, the fitted occupancy targets $P(\text{present},\ \text{key} \neq \text{null} \mid \text{block})$ instead of $P(\text{present} \mid \text{block})$, a bias by the factor $P(\text{key} \neq \text{null} \mid \text{block})$ that varies across blocks. Treating null as one more level of the extra key's distribution, $P(\text{segment or null} \mid \text{block})$ with the prior of Proposition 19, removes it.*

Proof. $P(\text{present}, \text{key} \neq \text{null}) = P(\text{present})P(\text{key} \neq \text{null} \mid \text{present}, \text{block})$. With null as a level the likelihood of each row factorises into the pair term and the extra-key term on all rows, so both are fitted on the full data. □

S8 found this: 84% of its values have no code, the earlier fit kept only coded values and generated 5,027 rows instead of 31,817. With the fix, 31,622 (check C26).

Proposition 27 (Nested refinement)**.** *When a fact is nested in a parent fact (values inside a filing, through a composite key), the parent-fact rows inherit their block from the parent model and are refined into sub-segments inside each block with the hierarchical prior $p_{k'l} \sim \text{Beta}(c\mu_{kl}, c(1-\mu_{kl}))$, $c = 2$, centred on the block's rate μ_{kl}. The collapsed score of a partition is a sum of per-segment Beta-binomial terms, so a merge of two sub-segments changes only their terms and merges in different blocks are independent.*

Proof. Given the block, sub-segments have independent rates under the prior; integrating them out gives a product over sub-segments and partner segments of $B(c\mu + S, c(1-\mu) + T - S)/B(c\mu, c(1-\mu))$. □

Without the centring, unrelated cells penalised every split and each block collapsed to one segment; with it, partial and complete filings are separated. *Check C27*: declared-not-filed and partial filings match within sampling error.

9.5 Dependence and posterior predictive draws

Proposition 28 (Rank likelihood with parameter expansion)**.** *Let* $\mathcal{D}(Y) = \{Z : z_{ij} < z_{i'j}$ *whenever* $y_{ij} < y_{i'j}\}$*. (a)* $P(Z \in \mathcal{D}(Y) \mid C)$ *depends on* Y *only through the within-column ranks, so the posterior of* C *is invariant to increasing transformations of any column. (b)* $\mathcal{D}$ *is invariant to positive rescaling of columns, so with* $\Sigma = DCD$ *(*D *diagonal) and* $\Sigma \sim \mathrm{IW}(p+1, I)$ *(uniform marginal priors on every correlation, Barnard, McCulloch and Meng 2000) the Gibbs sampler* $Z \mid \Sigma$ *(truncated normals within the rank bounds),* $\Sigma \mid Z \sim \mathrm{IW}(p+1+n, I + Z^\top Z)$ *has the posterior of* C *as the law of* $D^{-1}\Sigma D^{-1}$ *(Hoff 2007). (c) Extending the chain in blocks until split* $\hat{R} \le 1.1$ *and discarding the first half of the post-burn-in draws keeps every draw a state of the same chain.*

Proof. (a) $\mathcal{D}(Y)$ is defined by order relations only. (b) $P(Z \in \mathcal{D} \mid \Sigma) = P(D^{-1}Z \in \mathcal{D} \mid C)$ because $D^{-1}Z \sim \mathcal{N}(0, C)$ and $D^{-1}\mathcal{D} = \mathcal{D}$; the IW update is conjugate for Z given Σ. (c) Run-length control does not change the transition kernel; stopping by a diagnostic can bias finite runs (Cowles, Roberts and Rosenthal 1999), which the half-discard and the R2 drift check bound in practice. □

The implementation starts from the normal scores of the bins' mid-ranks, so the whole chain is a function of the bin order: *check C25* finds the draws bit-identical after a strictly increasing transform of every column, error 0.034 against the true correlations, $\hat{R} = 1.00$; on the real submissions table $\hat{R}$ fell from 1.58 to below 1.1.

Proposition 29 (Rare columns kept independent)**.** *Let column* X *take a value other than its mode with probability* q*. For any* Y*,* $\mathrm{TV}(P_{XY}, P_X \otimes P_Y) \le q(2-q) \le 2q$*. Hence sampling* X *independently of the other columns (given the segment) when fewer than* m *rows are non-modal (*$\hat{q} < m/n$*, default* $m = 5$*) changes every joint law involving* X *by at most* $2m/n$ *in total variation, while the copula correlations of such a column are not identified by the data.*

Proof. $\mathrm{TV} = \frac{1}{2}\sum_x P(x)\|P_{Y|x} - P_Y\|_1$. For the mode x_0, $P_{Y|x_0} - P_Y = \sum_{x \ne x_0} P(x)(P_{Y|x_0} - P_{Y|x})$ has norm at most $2q$; for the other values the norm is at most 2. So $\mathrm{TV} \le \frac{1}{2}[(1-q)2q + 2q] = q(2-q)$. □

Check C34: the resubmission flag of S8 (3 rows) is sampled independently; $\hat{R} = 0.99$.

Proposition 30 (Posterior predictive draws)**.** *Given the fitted partitions,* `redraw()` *draws all structural parameters (segment weights, block, occupancy and activity probabilities, multiplicity rates) from their conditional posteriors; sampling after each redraw gives draws from the posterior predictive distribution of table sizes and link counts conditional on the partitions. Posterior predictive p-values are conservative (concentrated near* $\frac{1}{2}$ *under the model), so extreme values signal misfit (Gelman, Meng and Stern 1996).*

Proof. Composition: $\theta \sim p(\theta \mid \mathrm{data}, z)$ then $D^* \sim p(D \mid \theta, z)$ gives $D^* \sim p(D \mid \mathrm{data}, z)$. □

Check C33: over 20 redraws of S8 the number of values has s.d. 498 and p-values 0.35 and 0.55 for values and reports.

9.6 Privacy on coarse tables

Proposition 31 (Distances with an atom at zero)**.** *When the real rows duplicate each other (a coarse table: a few low-cardinality columns), the leave-one-out distance distribution is* $a_r\delta_0 + (1 - a_r)G_r$ *with* $a_r > 0.05$*, its 5th percentile is 0 and the ratio of 5th percentiles is undefined. Writing the synthetic distances as* $a_s\delta_0 + (1-a_s)G_s$*, the pair of tests (i) exact-match rate* $a_s \le \max(0.01, 2a_r)$ *and (ii) 5th-percentile ratio of* G_s *to* G_r *at least 0.5 compares both parts of the mixture and reduces to the original test when* a_r *is small.*

Proof. A mixture law is determined by its weight and its components; each test compares one of them. For $a_r < 0.05$ the 5th percentile is attained in G_r and the original ratio applies. □

The earlier code compared synthetic distances including the atom with the real positive part only, and so failed every coarse table (S8 reports: ratio 0 although the exact-match rate was within its limit).

9.7 Defaults

$\pi_s = 0.9$ for occupancy, activity, null-rate and category-level priors, i.e. a combination that is absent from the data is a priori more likely to be forbidden than rare (`structure.structural_prior`). Copula mode `rank` with chain extension up to 600 iterations, $\hat{R}$ target 1.1. Rare columns: $m = 5$. Grain inference in `introspect`: a pair of foreign keys without duplicates where the chance-expected number of repeats $n(n-1)/(2 \cdot \text{space})$ is at least 10.

10 Small operational databases (v0.4)

The second real database was a small operational schema (11 tables, 15,607 rows: artists, albums, tracks, genres, media types, playlists, customers, employees, invoices and invoice lines; the catalog declares every key). It exercises situations that neither the fake scenarios nor the reporting model did: counts per parent that sit on a few fixed values (every customer has exactly 7 invoices; invoices have 1, 2, 4, 6, 9 or 14 lines), hierarchies of categorical columns (city, state, country), a hierarchy of 8 people of whom only 3 serve customers, price points (0.99, 1.99), long-tail labels (composers), a genre list just above the stratification limit, empty reference rows (playlists without tracks) and parents that always have children (every album has a track). Scenario S9 (a small shop with known truth) reproduces each of them; checks C35–C41 test the results below on it.

10.1 Rows per parent

Proposition 32 (Count-model evidence)**.** *Let $t_1, \dots, t_n$ be the counts of one segment. Under* $\mathrm{DP}(a, G_0)$ *with* $G_0 = \mathrm{Poisson}(\bar{t})$ *the marginal is the Pólya-urn product*

$$p_{\mathrm{DP}}(t_{1:n}) = \prod_{i=1}^{n} \frac{a\, g_0(t_i) + \#\{j < i : t_j = t_i\}}{a + i - 1},$$

which does not depend on the order of the counts. The engine compares $\log p_{\mathrm{DP}} - \frac{1}{2}\log n$ *with the Schwarz scores of Poisson (*$\frac{1}{2}\log n$*) and negative binomial (*$\log n$*) at their maximum likelihood, summed over segments, and uses the DP (posterior predictive: Bayesian bootstrap of the observed counts with mass* $a/(n+a)$ *on* G_0*) when the DP wins by a factor of at least 100. (a) If the truth* F *has finite support and is not in the parametric family,* $\frac{1}{n}\log \mathrm{BF} \to \mathrm{KL}(F \| G^\star) > 0$ *for the best parametric* $G^\star$*, so the DP is chosen with probability* $\to 1$*. (b) Under a parametric truth every first occurrence of a value costs the DP* $\log\{(a+i-1)/(a\, g_0(t_i))\}$*, i.e. about* $\log n$ *per distinct value, while later occurrences approach the empirical frequency; the parametric score pays a fixed* $\frac{d}{2}\log n$*. The DP is then chosen only when the counts repeat a small set of values far more than any Poisson or NB mixture allows.*

Proof. The urn product is the Blackwell–MacQueen joint law of an exchangeable sequence, hence symmetric. (a) The urn predictive converges to F (Ferguson 1973), so $\frac{1}{n}\log p_{\mathrm{DP}} \to \mathbb{E}_F \log F$; the parametric maximum likelihood gives $\frac{1}{n}\log \to \mathbb{E}_F \log G^\star$; penalties are $O(\log n)$. (b) Term by term from the urn product. □

Check C35: in 70 Poisson, NB and two-segment data sets the DP is never chosen; in 20 sets with counts on $\{1, 2, 4, 6, 9\}$ or constant it always is; the marginal is invariant under permutation. On the real database: lines per invoice $\log \mathrm{BF} = 254$, invoices per customer 101 (DP chosen); albums per artist -17, tracks per album -92 (not chosen).

Proposition 33 (Counts conditioned on a segment total)**.** *Let M parents of one segment receive R children in total (from the block model, which keeps how child segments choose parent segments). Drawing the parents' counts from $\pi(N) \propto \prod_i f(N_i)\,\mathbf{1}\{\sum_i N_i = R\}$, with f the DP predictive of Proposition 32, by repeatedly picking random disjoint pairs (i, j) and redrawing N_i from $f(x)f(s-x)$, $x = 0..s$, with $s = N_i + N_j$, leaves π invariant and is irreducible; the chain converges to π from any start. When f is Poisson, π is the multinomial $(R; 1/M, \ldots, 1/M)$. For a fact with two parents, drawing rows per anchor from the DP predictive and then each row's partner segment with probability $\propto \eta^*_{kl} W_{2l}$ changes only the law of the anchor total: under the Poisson block model, given the total, the split over partner segments is exactly this multinomial (Proposition 6), and the scale rule is kept.*

Proof. $\pi(N_i, N_j \mid N_{-ij}) \propto f(N_i)f(N_j)\mathbf{1}\{N_i + N_j = s\}$ is the exact full conditional of the pair, so each update is a Gibbs step. $f > 0$ on $0..R$ because of the mass $a/(n + a)$ on the Poisson base, so any allocation reaches any other by moving one unit at a time between pairs. Independent Poisson variables with equal means, conditioned on their sum, are multinomial with equal cells. □

Check C36: with a Poisson f, N_1 over 4000 draws has TVD 0.019 from Binomial$(10, \frac{1}{4})$; every draw sums to R; with observed counts all 7, every count is 7.

Proposition 34 (Mandatory participation)**.** *If every one of n parents has at least one child while the fitted Poisson rates leave $E_0 = \sum_i e^{-\mu_i} \geq 3$ parents empty in expectation, the Poisson model gives the observed "no empty parent" probability $\prod_i (1 - e^{-\mu_i}) \leq e^{-E_0} \leq e^{-3} \approx 0.05$. The engine then uses the shifted model $N = 1 + M$: for links between entities, each parent of a segment first receives one child (while the segment's rows last), the rest follow the usual weights; for facts, rows per anchor are $1 + \text{Poisson}(\max(\mu - 1, 0))$, which keeps the mean.*

Proof. $1 - x \leq e^{-x}$. Given a total $R \geq M$, assigning one child to each parent and the remaining $R - M$ by the multinomial of the weights is the conditional law of $1 + M_i$ with M multinomial. □

Proposition 35 (Copied anchors keep their activity)**.** *A copied table reappears row for row, each row its own segment. Whether such a row takes part in an occupancy fact at all (a playlist with no tracks) is a property of that known unit, so its activity is set to the observed indicator rather than drawn from a Beta posterior based on one observation; its pairs are still drawn from the block probabilities.*

Proof. Conditioning on the unit's identity: the posterior predictive of an observed unit's own fixed attribute is a point mass at its observed value. A Beta$(a + A, b + 1 - A)$ draw from a single observation would make an empty playlist active with probability about $a/(a + b + 1)$, and its tracks would then come from the prior (the shop data: 2140 rows in a playlist that has none). □

10.2 Hierarchies of categorical columns

Proposition 36 (Dependency groups)**.** *For categorical columns a, b (null a level of its own) let M_0: b i.i.d. Dirichlet(1)-categorical, and M_1: the first row of each level of a draws b from the same law and every later row repeats that value with probability $1 - \varepsilon$ ($\varepsilon = 10^{-3}$). (i) If a has no repeated level, $\log \text{BF} = 0$, so a near-unique column never determines another. (ii) If b is independent of a, each repeated row agrees with probability $\sum_l p_l^2 < 1$ and $\log \text{BF} \to -\infty$ linearly in the number of repeats. (iii) Columns linked by decisive dependencies are joined into one categorical over their observed combinations only if the combinations are no more than the levels of the finest member (plus $\lceil 10^{-3} n \rceil$ exceptions). Then the joint releases the information of that one column, every synthetic row satisfies the dependency, and no unseen combination (a city in the wrong country) can be generated.*

Proof. (i) Without repeats M_1 and M_0 assign the same Dirichlet-multinomial marginal to the same values. (ii) Each repeat contributes $\log(1 - \varepsilon) - \log p_b$ when it agrees and $\log\{\varepsilon/(L - 1)\} - \log p_b$

otherwise; under independence the expected contribution is $\sum_l p_l\{p_l \log(1-\varepsilon) + (1-p_l)\log(\varepsilon/(L-1))\} - \sum_l p_l \log p_l$, which is negative for $\varepsilon \le 10^{-3}$ unless b is nearly constant. (iii) The joint codec's levels are the observed combinations, in bijection with the finest member's levels when the dependencies are exact; its Dirichlet posterior puts mass only on them. □

Check C37: on the shop data the group (city, state, country) is found, every synthetic customer's state and country follow its city, the independent tier stays out, $\log \mathrm{BF} = -4851$ for independent columns and 0 for a unique determinant. Distance-to-closest-record keeps one column per group (the others add no information), so a table whose only attributes are one geographic hierarchy is not flagged as copied.

10.3 Tiny parents

Proposition 37 (Strata from an own column)**.** *An entity with fewer than* $2\times$`min_rows_per_segment` *rows has one latent segment, so it cannot learn which of its rows receive children. For such an entity, each categorical column c with $2 \le L_c < n$ levels is scored by $\sum_r \max\{0, \log \mathrm{BF}_r(c)\} - \log(\#\textit{candidates} \times \#\textit{relations})$, where BF_r compares a Poisson–Gamma model of children per row in relation r with one rate per level of c against one pooled rate (Gamma$(\frac{1}{2}, \frac{1}{2}/\bar{y})$ prior). If the best score exceeds $\log 100$, the levels of c become strata with one segment each, c is set from the segment, and the table is generated with its observed stratum counts (scaled). Children then go only to strata that had children, up to the structural prior (Proposition 17), and generating with fixed stratum counts is exact conditioning on the sufficient statistic of the composition.*

Proof. The Poisson–Gamma marginal is $\frac{b^a}{\Gamma(a)} \frac{\Gamma(a+\sum y)}{(b+n)^{a+\sum y}} \prod \frac{1}{y!}$; the BF is a ratio of such marginals. A relation the levels do not explain is given score 0, not a negative one, because the stratified model contains the pooled one for that relation (equal rates across strata). Given the counts, all arrangements of an exchangeable categorical sample are equally likely, so a uniform permutation of the scaled counts is the conditional law. □

Proposition 38 (Hierarchy of a given size)**.** *Every non-hierarchical entity is generated with exactly* $\mathrm{round}(s\,n)$ *rows; a multi-type Galton–Watson forest has a random size whose relative spread is large when the forest has few roots (one root: the size is a geometric-like variable). Repeating the branching process until $|T| = \mathrm{round}(s\,n)$ (within 1% for large forests, and with the stratum counts of Proposition 37 for tiny ones) draws from the process conditioned on its size, by rejection.*

Proof. Rejection with acceptance indicator $\mathbf{1}\{|T| = t\}$ samples the conditional law (Proposition 14); the expected number of attempts is $1/P(|T| = t)$. When no attempt in the budget is accepted, the closest draw is used and the deviation is reported. □

Check C38: the shop's staff (8 rows) is stratified by office (agents' branch), all synthetic customers are served by sales agents, the staff has 8 rows with the source's stratum counts, one root and no cycle. *Check C10* still compares the size formula with the unconditioned process.

10.4 Stratification by larger code lists

Proposition 39 (Code lists that explain links)**.** *A copied parent with more levels than* `stratify_max_levels` *is used as a stratifier when its levels explain how many children the entity's rows have (Poisson–Gamma, as above) or which partners they meet in two-parent facts: a symmetric Dirichlet-multinomial over partner rows per level against one pooled, each relation contributing* $\max\{0, \log \mathrm{BF}\}$*, minus* $\log(\#\textit{relations})$*, above* $\log 100$*. Segments are nested in the strata, so a block of a fact that pairs a partner row with a segment of one genre never generates tracks of another genre.*

Proof. The Dirichlet-multinomial marginal is exact for exchangeable partner draws within a level; the comparison is a ratio of marginals. Nesting: Proposition 25. □

Check C39: the shop's playlists hold only tracks of their own genres; with genre (30 levels) as stratum the share of synthetic playlist rows with a genre the playlist never has is at most 2% (82% before).

10.5 Values

Proposition 40 (Tail levels and few-valued numbers)**.** *(a) A categorical with more than* `max_levels` *levels models the rarest as one level. Decoding that level by a Bayesian bootstrap over the tail levels seen in at least k rows (default 5) gives each of them the marginal of the full Dirichlet posterior restricted to those levels; levels seen in fewer than k rows are never released, and the placeholder is never written. (b) An integer or decimal column with at most 30 distinct values is modelled as an ordered categorical on those values, so no value between them is generated.*

Proof. (a) Aggregation property: if $(p_1, \dots, p_K, p_O) \sim \mathrm{Dir}(\alpha_1, \dots, \alpha_K, \sum_{j \in O} \alpha_j)$ and, independently, $w \sim \mathrm{Dir}(\alpha_j, j \in O)$, then $(p_1, \dots, p_K, p_O w) \sim \mathrm{Dir}(\alpha_1, \dots, \alpha_K, \alpha_j, j \in O)$. (b) A continuous Bayesian-bootstrap marginal interpolates its quantile knots, so it puts positive mass between two atoms (the shop's prices came out as 1.54 or 4.26); a categorical marginal is supported on its levels. □

Aggregates are rounded to the precision of the stored column (a sum of prices is stored to the cent); without this, totals 5.9399 … and 5.94 count as different values in every comparison. *Checks C39, C41*: lines per invoice stay on $\{1, 2, 4, 6, 9\}$, every customer has 5 invoices, totals equal the sum of their lines to the cent, prices are 0.99 or 1.99, the empty playlist stays empty, the shop scorecard passes; a categorical's tail decodes only to levels seen in at least 5 rows.

Labels. The text model removes every token seen in fewer than k rows (Section 7). For names and titles most words are unique to their row; when fewer than half of the rows keep a word after that removal, the model could only recombine punctuation and filler words (", &"), so it generates surrogate labels ("Artist 04217") instead; otherwise generated texts must contain a word and may not open or close on punctuation or leave a bracket open (a rejection step, as in Proposition 14). Identifiers without a format shared by 5 rows (company names, street addresses) are surrogate labels too, and person-name columns give first or last names according to the column. *Check C41.*

Proposition 41 (Evidence for a grain)**.** *Let a table have n rows whose columns (A, B) are drawn independently with level probabilities p_a, q_b. The number of row pairs that agree on (A, B) has mean $\lambda = \binom{n}{2} \sum_a p_a^2 \sum_b q_b^2$, and $\hat{\lambda} = \binom{n}{2} \hat{S}_A \hat{S}_B$ with $\hat{S}_A = \sum_a c_a(c_a - 1)/(n(n-1))$ estimates it; each factor is unbiased for its sum of squares. When the probabilities are small, the count is approximately Poisson(λ) (Chen–Stein), so a table with no repeated pair gives a likelihood ratio of about e^{λ} for "each pair occurs at most once" against independence. The doctor proposes a unique key only when $\hat{\lambda} \geq 10$ ($e^{-10} < 5 \cdot 10^{-5}$). Below that, regular counts are reported as a property of the source and nothing is changed.*

Proof. $\mathbb{E}[c_a(c_a - 1)] = n(n-1)p_a^2$ for a multinomial count, which gives unbiasedness. Two rows agree with probability $\sum_{a,b}(p_a q_b)^2 = \sum p_a^2 \sum q_b^2$, so linearity over the $\binom{n}{2}$ pairs gives λ. The Poisson limit for the number of matching pairs (the birthday problem) is Theorem 1 of [5]: the total variation distance is at most $\min(1, 1/\lambda)\,(b_1 + b_2)$, and both terms vanish when $\max_{a,b} p_a q_b \to 0$. *Check C42* (pair count within 0.3% of Monte Carlo, plug-in within 0.2%, $P(0)$ within 0.01 of $e^{-\lambda}$). □

On the music store, lines of an invoice never repeat a track, but with 2240 lines spread over 3503 tracks only $\hat{\lambda} = 0.9$ repeats were expected, so this is not evidence of a rule; on the playlists, $\hat{\lambda} = 2241$ and the grain is real.

10.6 Introspection

Type inference now uses the table's context: a bare `name` is a person's name only in a table with personal columns; near-unique labels of a small table that is not a party (genres, media types) are reference labels; near-unique labels of a party table (reporters, customers, staff: a lexicon of party nouns) are identifiers; other near-unique or long-tail multi-word labels (titles, composers: at least 20% of rows hold values seen fewer than 5 times) are short text. Columns equal to a parent's column on 99.9% of rows (null equal to null) become `lookup` columns; parent columns equal to a count, a sum or a sum of products of child columns become `aggregate` rules; the anchor of a fact is the parent whose name the fact extends and whose keys it covers. A column named like a key whose parent the names do not reveal is listed as a question (surrogate keys of most tables share the range $1..n$, so values alone cannot decide). *Check C40.*

10.7 Defaults (v0.4)

`facts.count_model_bf` 100; `marginals.fd_bf` 100; `segments.attribute_strata` `auto` (a table name to column map forces one); mandatory participation at $E_0 \geq 3$; tree size tolerance 1%.

11 Data warehouses with order headers (v0.5)

The third real database was a sales data warehouse (AdventureWorks DW: 27 tables, 1.05 million rows; composite keys, a composite foreign key into a fact, order numbers stored on the lines, three role-playing date keys per order, amounts that are identities of other columns, a running stock balance). Its first two runs exposed 38 findings (report: `realdata/adventureworks/README.md`). This section states the model changes that remove their causes and the properties they rely on. Scenario S10 (a sales warehouse with known truth) reproduces each structure; checks C43–C45 test them.

11.1 Identifying the data model

A drafted model is a set of dependencies found in samples: unique column combinations (keys), inclusion dependencies (links), functional dependencies (lookups, order headers), algebraic constraints (derive rules) and order dependencies (running balances). Each candidate is accepted only by the following test.

Proposition 42 (Significance of a discovered dependency)**.** *Let a candidate dependency hold on s of n eligible sampled rows, and let p_0 be the probability that it holds on one row by chance: the collision probability $\sum_v p_v^2$ of the column for an equality with another column, $\sum_v p_v^m$ over the value shares of a determined column for a functional dependency checked on groups of m rows, and the agreement rate with a randomly permuted input for algebraic and cumulative rules. Introspect accepts the candidate iff $n \geq 30$, $s/n \geq 0.999$ and $P(\mathrm{Bin}(n, p_0) \geq s) \leq \alpha = 10^{-9}$. With M candidates tested, the probability of accepting any dependency that holds only by chance is at most $M\alpha$ (Bonferroni); on AdventureWorks $M < 10^5$, so at most 10^{-4}. A rule that is true on the whole table is accepted with probability $\to 1$ as soon as $n\,\mathrm{KL}(1\|p_0) = -n \log p_0 > \log(1/\alpha) + O(\log n)$, i.e. about $21/(-\log p_0)$ rows.*

Proof. Under the null each row agrees independently with probability at most p_0, so the binomial tail bounds the false-acceptance probability of one test; the union bound gives $M\alpha$. If the rule holds on every row, $s = n$ and the tail is p_0^n, which is $\leq \alpha$ iff $n \geq \log(1/\alpha)/(-\log p_0)$. □

The fixed threshold used before (chance share below 0.5) rejected true rules on skewed columns (89% of reseller lines carry no discount, so the discount-rate lookup has $p_0 = 0.89$; with $n = 20{,}000$ the tail is 10^{-1000}) and accepted chance agreements on small samples. Samples are simple random samples drawn with a seeded generator from the table's primary key (Section 11, C44: two runs

give the same spec); groups (orders, products) are read whole as cluster samples for within-group rules. Personal and free-text columns are candidates for lookups too: a billing address equal to the customer's address on every row is a functional dependency like any other, and generating it as a lookup copies the synthetic customer's generated address, which keeps both the identity and the privacy (an earlier v0.5 draft excluded them and lost this rule on Chinook).

Proposition 43 (Order headers: lossless split)**.** *Let a fact F with a document number d satisfy the functional dependency $d \to H$ for a set of header columns H. Then $F = \pi_{d,H}(F) \bowtie \pi_{d,L}(F)$ for the remaining columns L (Heath 1971). Generating $\pi_{d,H}$ as an entity (one row per document, with its own links) and the lines as a fact anchored on it, then joining, gives synthetic data in which every document has exactly one header and line numbers $1..n$ by construction; the scale rule applies to documents per customer and lines per document.*

Proof. Heath's theorem: a decomposition along a functional dependency whose left side is a key of one component is lossless. The synthetic header table has one row per document by construction, so the join repeats it on each line. □

Proposition 44 (Offset keys)**.** *Let b and k be keys into one parent whose order column o is injective (a calendar). The map $(b, k) \mapsto (b,\, o(k) - o(b))$ is a bijection from pairs of parent rows onto pairs (row, attainable gap). Modelling the gap as an attribute therefore loses nothing, keeps the gap distribution (ship = order +7 days on 100% of rows) exactly, and a draw whose $o(b) + \text{gap}$ is not an order value of the parent is outside the support of the source; redrawing it (Proposition 45) gives the model conditioned on the support.*

Proof. Given b, $k \mapsto o(k) - o(b)$ is injective because o is; its inverse is the lookup of $o(b) + \text{gap}$. □

11.2 Generating dependent records

Proposition 45 (Redraw instead of delete)**.** *Let the rows of a fact be independent given their anchors, row $i \sim Q(\cdot \mid a_i)$, and let V be the set of valid rows (rules, unique keys, offsets landing on a parent row). The law that keeps the number of rows per anchor and puts all mass on valid rows, closest to Q in Kullback–Leibler divergence, is $\prod_i Q(\cdot \mid a_i, V)$, and redrawing each invalid row from $Q(\cdot \mid a_i)$ until it is valid samples it exactly. Deleting invalid rows instead conditions the counts as well: a customer's expected number of kept rows is multiplied by $Q(V \mid a_i)$.*

Proof. For a product reference measure and the constraint that each factor lives on V, the I-projection factorises and each factor is $Q(\cdot \mid a_i)$ restricted to V and renormalised (Csiszár 1975). Rejection sampling from $Q(\cdot \mid a_i)$ returns exactly that restriction. Under deletion the number of kept rows of anchor i is binomial with success probability $Q(V \mid a_i)$. □

On AdventureWorks deletion removed 57,750 of 60,398 internet sales (a unique key over an order number drawn as a 200-level category) and 46,224 of 60,855 reseller sales (date checks over independently drawn keys). Both causes are now modelled (Propositions 43, 44); leftovers after the last round are deleted, counted and scored (`rows_removed_share` ≤ 0.01, doctor D23).

Proposition 46 (Shares of copied rows)**.** *Rows d of a copied table in segment k receive c_d links. Model their rates as $\lambda_d \sim \text{Gamma}(a_k, \beta_k)$, $c_d \mid \lambda_d \sim \text{Pois}(\lambda_d)$, with a_k estimated by moments, $a_k = \bar{c}^2/(s^2 - \bar{c})$ when $s^2 > \bar{c}$ and $a_k = \infty$ otherwise. Drawing each row's share from the posterior $\text{Gamma}(c_d + a_k)$ and the synthetic counts from the resulting Poisson gives counts whose marginal law over rows equals the source's (negative binomial with the fitted a_k). The previous plug-in, shares $\propto \text{Gamma}(c_d + 1/m_k)$, adds the source's Poisson noise a second time: on a calendar with 4.6 orders per day the variance/mean of orders per day became about 2 instead of 1.*

Proof. If $\lambda \sim \pi$ and $c \mid \lambda \sim p(c \mid \lambda)$, then a draw $\lambda^* \sim \pi(\cdot \mid c)$ is again distributed as π (the data-averaged posterior equals the prior), so $c^* \mid \lambda^*$ has the marginal law of c. With $a = 1/m$ the posterior mean of λ is about c, and $c^* \sim \text{Pois}(\approx c)$ has variance $\text{Var}(c) + \mathbb{E}c$. □

Proposition 47 (Per-row segments need data per row)**.** *If a copied table with m rows keeps one segment per row, every segment k of an entity linked to it carries m free rates $\eta_{k\cdot}$. Splitting one segment into two adds m rates; under the Gamma prior each costs an Occam factor of order $\frac{1}{2}\log(1+n_{kd}/\beta)$ in the collapsed marginal, so a split that the data support by a log-likelihood gain G is rejected when $G < \frac{m}{2}\overline{\log(1+n_{kd}/\beta)}$. With few links per row this blocks every split of the other side (S10: the products collapsed from four categories to two segments, and the price–category and buyer–category dependences were lost). Rows are therefore kept as segments only when each relation carries at least* `min_rows_per_segment` *links per row; otherwise the copied table is clustered like any entity and its rows keep their own shares (Proposition 46): a degree-corrected block model (Karrer and Newman 2011), whose cost per extra segment is K_{copied} rates instead of m.*

Proof. Laplace approximation of the Gamma–Poisson marginal of one block: $\log p(C \mid E) = \log p(C \mid \hat{\eta}) - \frac{1}{2}\log(1+E/\beta) + O(1)$. □

Proposition 48 (Tied header segments)**.** *Let the header entity H take its parent's segment, $z_j = z_{p(j)}$. The Gibbs full conditional of a parent row i is*

$$P(z_i = k \mid \cdot) \propto \pi_k \, f(x_i \mid k) \prod_{r \ni i} L_r(i,k) \; \text{Pois}(m_i \mid \mu_k) \prod_{j : p(j) = i} f(x_j \mid k) \prod_{r \ni j} L_r(j,k),$$

which the sampler evaluates by summing the children's row log-likelihoods into their parent's row. The collapsed merge score of a parent partition adds the children's attribute and relation marginals under the induced partition and no partition prior of their own. The model of lines given the customer is then the one the unsplit fact had (rates per customer segment and product segment), while documents keep their own header columns and line counts.

Proof. The children's segments are a deterministic function of z_i, so changing z_i changes every factor that involves z_j, $p(j) = i$; all other factors do not depend on z_i. The partition of the children is not a free variable, so it has no prior. □

Without the tie the header segments collapsed to one or two (about two lines per order give little evidence per order) and product choice stopped depending on the customer (S10: share of bikes in segment C orders 0.24 instead of 0.03; with the tie 0.07; check C43).

Proposition 49 (Running balances)**.** *A cumulative rule $y_t = y_{t-1} + s_t$ within groups is generated from the steps. If every source group opens at the same value ($y_1 - s_1 = c$ on 99.9% of groups) the synthetic groups open at c. If the source balance is never negative and the step subtracts a column ($s = a - b$), the subtracted column is reduced to $\min(b,\, y_{t-1} + a - \mathit{floor})$, the rule of an account that cannot pay out more than it holds; then $y_t \geq \mathit{floor}$ on every row and the identity still holds.*

Proof. Induction over t: if $y_{t-1} \geq \text{floor}$ and $y_{t-1} + a - b < \text{floor}$, the reduced $b' = y_{t-1} + a - \text{floor} \in [0, b)$ gives $y_t = \text{floor}$. □

11.3 Run time of the segment merge

Proposition 50 (Merge from block statistics)**.** *Every term of the collapsed merge score (attributes, Poisson and occupancy blocks, counts, trees, tied children) depends on the partition z of the merged entity only through statistics that are sums over rows within segments. Merging segments a, b adds row b of each statistic to row a along the axes indexed by that entity. Scoring a candidate therefore costs $O(K \cdot K_{\text{other}})$ per relation instead of a pass over its rows, and gives the same value (checked to 2×10^{-16} relative on S4, S7–S10).*

Proof. Block counts, sums of measures and segment sizes are additive over the rows of a segment; the remaining factors (e.g. $E = n_1 n_2^\top$, $T = A\, m_2^\top$) are products of additive statistics and are formed after merging. □

Proposition 51 (Merge gain from two slices)**.** *Call a term of the merge score separable in entity e if it equals $\sum_k g(S_{k\cdot}) + h(\sum_k S_{k\cdot})$, where $S_{k\cdot}$ is the slice of its statistics for segment k of e. Attribute terms, segment sizes, link, single-count, tied-count and occupancy terms (without a per-stratum prior) are separable; tree terms and links of an entity to itself are not. For a separable term the gain of merging a and b is $g(S_{a\cdot} + S_{b\cdot}) - g(S_{a\cdot}) - g(S_{b\cdot})$, i.e. the term evaluated on the two slices merged minus on the two slices apart, at cost $O(K_{\text{other}})$ instead of $O(K\,K_{\text{other}})$.*

Proof. Merging changes only rows a, b of each sliced statistic, and leaves their column sums, hence h, unchanged. □

Checked against full rescoring on 537 candidates of five scenarios and every entity of AdventureWorks (largest relative difference 1.5×10^{-10}, same best pair everywhere). On AdventureWorks the merge took more than 8 minutes when rescoring over rows (stopped), 276 s from block statistics, and 84 s with two-slice gains; the customer table carries 709 segments because it is stratified by its 655 geographies.

11.4 Structural zeros, mandatory participation and hierarchies

Proposition 52 (Structural zeros in an occupancy fact)**.** *Let a nested fact's key require that a partner column equals a column of its anchor (a cell's template equals the filing's template). Pairs with different values are impossible. In the occupancy model put group codes g_i on anchors and h_j on partners and replace every count of pairs at risk by eligible pairs: $T_{kl} = \sum_g N^{(1)}_{kg} N^{(2)}_{lg}$, and the non-present pairs of row i by the partners of its own group. This is the quasi-independence model of a table with structural zeros (Goodman 1968). Sampling only eligible pairs then needs no deletion, and $\mathbb{E}$[rows per anchor] equals the source's within each block, whereas the full-grid model underestimates p_{kl} by the share of ineligible pairs and loses those rows when they are deleted.*

Proof. Under the full-grid model $\hat{p}_{kl} = E_{kl}/(A_k m_l)$, while only $T_{kl} \le A_k m_l$ pairs can occur; drawing on the eligible pairs with $\hat{p} = E/T$ gives $\mathbb{E} = E$ in every block. The statistics $N^{(1)}$, $N^{(2)}$ are additive over segments, so Proposition 50 still applies. □

On the bi schema 784–1,088 reported values (1.2–1.6%) were deleted in every earlier version; now none are.

Proposition 53 (Mandatory participation in an occupancy fact)**.** *If every source anchor has at least one row (an order header always has lines) and the fitted model expects at least 3 anchors without pairs, $\sum_i \prod_l (1 - p_{z_i l})^{m_l} \ge 3$, the pairs of each active anchor are drawn conditioned on at least one being present, by redrawing empty anchors (rejection) and, for anchors still empty after 20 rounds, by the sequential method below; both are exact for the zero-truncated law. The rates are then fitted for that law: in segment k the free rates ($0 < p_{kl} < 1$) are scaled by the c_k that solves*

$$\sum_{i:z_i=k} \frac{\mu_i(c_k)}{1 - P_{0,i}(c_k)} = \sum_{i:z_i=k} n_i, \qquad \mu_i(c) = \sum_l c\,p_{kl} m_l, \quad P_{0,i}(c) = \prod_l (1 - c\,p_{kl})^{m_l},$$

with n_i the observed pairs of anchor i (m_l replaced by the anchor's eligible partners under Proposition 52).

Proof. For any count $N \ge 0$, $\mathbb{E}[N \mid N \ge 1] = \mathbb{E}[N\,\mathbf{1}\{N \ge 1\}]/P(N \ge 1) = \mathbb{E}[N]/(1 - P_0)$, exactly for the Poisson-binomial count of an anchor's pairs. The left side increases in c (the zero-truncated mean of a sum of Bernoullis increases with each success probability), so the root is unique and found by bisection. This is the method-of-moments estimator of the truncated model; in the Poisson limit it is the maximum-likelihood estimator of the zero-truncated Poisson. Rates fitted without it give $\mathbb{E}[N \mid N \ge 1] = \bar{n}/(1 - P_0)$, too high by the factor $1/(1 - P_0)$. □

Sequential draw. With the partner blocks in a fixed order and $P_{0l} = (1-p_l)^{m_l}$, the first non-empty block is l^* with probability $(1 - P_{0l^*}) \prod_{l<l^*} P_{0l}/(1 - P_0)$, its count is binomial truncated at zero, later blocks are unconditioned binomials, and partners are uniform within blocks. This factorises the conditional law exactly, and its cost does not depend on P_0; rejection alone needs $1/(1 - P_0)$ rounds, unbounded when an anchor segment always has exactly one pair (then $c_k \to 0$ and the law tends to one pair with block probabilities $\propto p_l m_l$). Without it about 15% of AdventureWorks internet orders received no line and vanished (23,565 orders instead of 27,659, 2.54 lines per order instead of 2.18), and with them 12% of the customers' only purchases. Conditioning alone restored the orders but gave 2.46 lines per order (11% too many lines, 67,994 instead of 60,398): the fitted rates already averaged the source's 2.18 lines over orders that all had lines. With the rates fitted for the truncated law the run gives 27,659 orders, 2.18 lines per order and 60,250 lines (source 60,398); the scale factors range from 0.000 (segments of one-line orders, drawn by the sequential method) to 2.4 (a value above 1 means the posterior draw of the rates sat below the segment's observed mean). On simulated data with known rates (Check C43) the uncorrected rates overstate the pairs per anchor by 20–25%, the corrected ones are within 4% (sampling noise) over 16 seeds.

Proposition 54 (Hierarchies generated level by level)**.** *Let a self-reference be a multi-type Galton–Watson process whose type is the node's segment and depth, with rates λ_{kd} and child-segment probabilities ψ_{kd}. Given the numbers $n_{k'd}$ of nodes of each type at depth d, the parents of the depth-d nodes of segment k' are i.i.d. among the depth-$(d-1)$ nodes with probabilities $\propto w_i \lambda_{z_i,d-1} \psi_{z_i,d-1}(k')$ (w_i the node's frailty). The generator therefore fixes* $\mathrm{round}(s\, n_{kd})$ *and draws the parents; size, depth profile and composition match the source at every scale.*

Proof. Independent Poisson counts conditioned on their total are multinomial with probabilities proportional to their means; children of type k' of node i are Poisson with mean $w_i \lambda_{z_i,d-1} \psi_{z_i,d-1}(k')$, independently over i. □

The rates by level have the prior $\mathrm{Gamma}(\lambda_k, 1)$ (one pseudo-node at the segment's rate). Before, the process was conditioned on its size only: on AdventureWorks a segment mixing managers and staff grew chains to depth 7 (source 4), and the 18-person sales branch (one vice-president, three managers, 14 representatives) vanished in most draws with one decision near the root; with it went the quotas of the sales staff (finding F38: 6 quota rows instead of 163). Level by level, three seeds give 17–18 sales representatives (source 17), and the scorecard of the employee table passes 58–59 of 59.

12 Normalised and denormalised warehouses (v0.6)

The fourth real database was the Contoso Data Generator V2 release (100k customers; 93,470 orders, 223,974 order lines; a denormalised sales table next to the normalised orders and order lines; exchange rates by date and currency pair with no links; geography stored on the customer row; prices that are list prices times a factor). Its test (from v0.5, unedited drafts) exposed findings K01–K18 (report: `realdata/contoso/README.md`). This section states the model changes that remove their causes and the properties they rely on. Scenario S11 (a retail warehouse with known truth) reproduces each structure; checks C47–C56 test them. Regressions on AdventureWorks and S11 exposed seven more (R1–R7), also fixed here.

12.1 Identifying the data model

Proposition 55 (Functional dependencies tested on rows)**.** *For columns x, y on n sampled rows let $s = \frac{1}{n} \sum_g \max_v n_{gv}$, the share of rows whose y is the most common y of their x-group g, and let p_0 be the mean of the same statistic over random permutations of y. Introspect accepts $x \to y$ iff $s \geq 0.999$, the binomial test of Proposition 42 passes with (s, n, p_0), and $\exp(-n(s - p_0)^2/2) \leq \alpha$. Under independence of x and y the probability of acceptance is at most α. Singleton groups agree*

under both hypotheses and add no evidence, so a near-unique x is never accepted, and a code with few large groups (8 countries) is tested on its thousands of rows. The group-based test used before counted groups, so a dependency from 8 countries had $n = 8 < 30$ and was never tested (the currency of the customer's country, K09).

Proof. Under independence every permutation of y is equally likely given the data, so the permutation law is the exact null of s, with mean p_0. A transposition of two y values changes s by at most $2/n$; the bounded-differences inequality for random permutations (McDiarmid 1989) gives $P(s - p_0 \geq t) \leq \exp(-2t^2/(n(2/n)^2)) = \exp(-nt^2/2)$. The binomial condition is kept as well (it is the stricter one when p_0 is close to 1). □

Proposition 56 (Implied dimensions)**.** *Let an entity E satisfy $k \to Y$ for a code column k and a set of columns Y. Then $E = \pi_{k,Y}(E) \bowtie \pi_{k,R}(E)$ (Heath 1971). Generating $D = \pi_{k,Y}(E)$ as a copied parent, k as a link of E to D and Y as lookups $k.y$ gives synthetic rows with $k \to Y$ exactly and every (k, Y) combination an observed one, at any number of levels; the expected number of synthetic rows per k follows the source degrees (Proposition 46). If the dependency holds on a share $1 - \varepsilon$ of the rows, using the first Y of each k changes the Y values of at most a share ε of the rows, so every marginal of Y moves by at most ε in total variation.*

Proof. Heath's theorem as in Proposition 43; the lookup reads D's row of each synthetic k, so Y is a function of k. A row whose Y differs from its group's first values is the only row whose value changes. □

Anchors. A code k drawn as a link carries the dependence of the other attributes on Y only through the segments. When an attribute depends on a coarse label of k (latitude on the country, 8 countries against 565 areas) the segments, fitted to the whole entity, need not resolve it: Contoso's synthetic customers had the same mean latitude in every country (K16). An anchor $a \in Y$ (with $k \to a$) is kept as a modelled attribute and k is drawn given it (the rule $k.a = a$ then holds on every row and is reported like any other).

Proposition 57 (Anchored dimensions)**.** *Let $a \in Y$ be the anchor and let the link k be drawn from the link model restricted to $\{k : D[k].a = a\}$: a parent segment t with probability $\propto \phi_{zt} W_t(a)$, where $W_t(a)$ is the share of segment t's rows with anchor a under the link's member law (degree weights, or uniform under a count model), then a member of t with anchor a by that law. Then (i) every synthetic row keeps $k \to Y$ and an observed (k, Y) combination (Proposition 56); (ii) the joint law of a with the other attributes is the attribute model's, as for any categorical; (iii) given a, k has the link model's law restricted to the codes of a. If the attribute model reproduces the source law of (a, X) and the link model reproduces $k \mid a$, the joint law of (k, Y, X) is reproduced whenever $X \perp k \mid a$ in the source.*

Proof. (i) Y is still read from $D[k]$, so the anchor column is overwritten by $D[k].a$, which equals the modelled a on every row. (ii) Only the link is drawn. (iii) $Q(k \mid z) = \sum_t \phi_{zt}\, \omega_t(k)$ with ω_t the member law of segment t; restricted to the codes of a this is $\sum_t \phi_{zt} W_t(a)\, \omega_t(k)/W_t(a)$, which the two stages sample. When an attribute rule makes the row's attributes be redrawn, its link is drawn again from the new anchor. The last claim is the factorisation $P(k, X) = \sum_a P(a, X)P(k \mid a)$ under $X \perp k \mid a$, with $Y = D[k]$. □

The Metropolis–Hastings kernel used first left 289 of 105,000 Contoso customers outside their country's areas after 200 repair rounds, because a segment's link model can give the anchor's areas no weight; the two-stage draw has no rejection, and a row whose segment gives the anchor no weight takes its segment from the link's overall mixture (logged; none in Contoso). Introspect picks as anchor the finest categorical label among Y with 2 to 50 levels (Country rather than Continent; SubCategoryName in Product), since a finer label keeps more of X's dependence while the bounded number of levels keeps the attribute model's cells populated.

A joint categorical over the observed combinations (Proposition 36) discloses the same combinations but pools those beyond 200 levels: with 565 states, 610 full state names and 8 countries the Contoso customers kept (Country, State)→StateFull on 19% of the synthetic rows (K13). Integer codes (SubCategoryKey) were not candidates of the joint categorical at all (subcategory→category 69%).

Proposition 58 (Labels nested in a code)**.** *Let a categorical column c have more levels than the marginal model keeps (200) and let a coarser column g of the row (a state, or a dimension code) hold most of each label's rows: on the sampled rows whose label occurs at least twice, the share s of rows in their label's most common g is at least 0.8 and passes the test of Proposition 55. Generate c after g from*

$$p(c \mid g) = \frac{n_{gc} + \beta\, p_G(c)}{n_g + \beta}, \qquad p_G(c) = \frac{n_c + 1}{n + V},$$

over the V labels held by at least m source rows (`text.min_token_rows`, default 3). (i) This is the posterior predictive of a Dirichlet(βp_G) prior on each group's label distribution, so as $n_g \to \infty$ it converges to the source's $P(c \mid g)$. (ii) No label held by fewer than m rows is ever generated. (iii) A label is generated outside its source groups with probability at most $\beta/(n_g + \beta)$ per row.

Proof. (i) Dirichlet–multinomial conjugacy; the prior weight $\beta/(n_g + \beta) \to 0$. (ii) Only vocabulary labels have mass. (iii) Outside g's labels only the prior term $\beta p_G/(n_g + \beta)$ has mass. □

Contoso's 34,581 cities were a categorical pooled to 200 levels and drawn without their state: City→State held on 0.92 of the source rows and 0.10 of the synthetic ones, and the Customer table's worst association (City with StateFull, 0.77 against 0.20) was the run's last scorecard failure (K17). The text model with `whole_values` (each label one token, order 0) is this predictive; its end-of-text token takes part of the base measure, so the effective prior weight is β times the labels' share of p_G, below β, and (i)–(iii) hold as stated. The labels lose their copula dependence on the row's other columns except through g, which is where the source keeps it. Check C53 tests the draft and the draws.

12.2 Columns that are functions of other columns

Proposition 59 (Ratio columns)**.** *Let $y = \mathrm{round}_d(xf)$ where f takes values in a finite set F of numbers with at most k_0 decimals and d is the stored precision of y. (a) On every row with $|x| \geq 10^{k_0 - d}$, $\mathrm{round}_{k_0}(y/x) = f$. Hence the coarsest $k \in \{2, \ldots, 8\}$ whose rounded ratios reproduce y to the stored precision on at least 99.9% of rows recovers F when the condition holds on those rows. (b) Modelling f as a categorical attribute and deriving $y = \mathrm{round}_d(xf)$ makes $y \in \mathrm{round}_d(xF)$ on every synthetic row, keeps the joint law of f with the other attributes that the attribute model gives any categorical, and keeps two ratio columns that share their factor ($y_1 = \mathrm{round}_d(x_1 f)$, $y_2 = \mathrm{round}_d(x_2 f)$) in the same proportion on every row. (c) Factors seen in fewer than 5 rows are replaced by the nearest common factor; y then changes by at most $|x|\,\Delta f$ on those rows, whose number is reported.*

Proof. (a) $|y - xf| \leq \frac{1}{2}10^{-d}$, so $|y/x - f| \leq \frac{1}{2}10^{-d}/|x| \leq \frac{1}{2}10^{-k_0}$ and rounding to k_0 decimals returns f (on a tie the stored value also matches the other neighbour). Any $k < k_0$ rounds some $f \in F$ away from itself and fails to reproduce the rows that carry it, unless those are fewer than 0.1%. (b) The derive rule is applied after the factor is generated; a categorical attribute is modelled like any other. (c) $|xf - xf'| = |x||f - f'|$. □

Discovery tests x among the row's numeric columns and its parents' (one hop), with at most 50 factors; the chance level is the share of rows reproduced by some factor of F when x is permuted. A column is never its own base's base. Contoso: unit price = list price × one of 16 factors, unit cost = cost × the same factor, net price = unit price × one of 15 discount factors, all to 4 decimals on 100% of rows; v0.5 drew the three prices from a copula (K14, 1% and 0.07% of rows right).

Proposition 60 (Column offsets)**.** *For two date (or number) columns a, b of a row the map $(a, b) \mapsto (a, b - a)$ is a bijection; modelling the gap $b - a$ as an attribute and deriving b keeps the gap distribution, its support (never negative) and its dependence on the other attributes as the attribute model gives them, and $b = a$ exactly on every row whose gap is 0. Discovery requires a gap that is never negative on 99.9% of rows with at most 60 distinct values, beyond the chance share of permuted pairs whose gap falls in that set, and a set on 99.9% of the rows where b is set ($b = a +$ gap is null wherever a is; AdventureWorks' product StartDate is always set while EndDate mostly is not).*

Proof. As Proposition 44 with the identity as the order column. □

Proposition 61 (Views, reference tables, keyed and mapped lookups)**.** *(a) Let V and B have the same key U with the same key values, and let every column of V be a column of B (equal on the joined rows) or a lookup through V's links. Then $V = B \bowtie$ (its parents); rebuilding V from the synthetic B and parents by the same lookups gives $V_{\text{syn}} = B_{\text{syn}} \bowtie (\text{parents}_{\text{syn}})$ exactly, with $|V_{\text{syn}}| = |B_{\text{syn}}|$. (b) A keyed lookup $y = R[v](e_1, \ldots, e_m)$ into a copied reference table R keyed by $(r_1, \ldots, r_m)$ holds on every synthetic row whose key $(e_1, \ldots, e_m)$ is a key of R; if R is complete over the product of the key domains and each e_j stays in the domain of r_j, it holds on every row. (c) A mapped lookup $y = m(p.x)$ holds on every row whose $p.x$ is in the map.*

Proof. (a) Every column of V is determined by B's row and the parents; the rebuilt rows are those values. (b), (c) The lookups are evaluated on the synthetic rows after the columns they read. □

Contoso's Sales table is OrderRows joined to Orders with ExchangeRate = rate(OrderDate, 'USD', CurrencyCode); CurrencyExchange is the complete grid of 4,018 dates × 5 × 5 currencies. v0.5 modelled Sales as a second fact (K04: 0% of its rows matched the order lines) and dropped CurrencyExchange (K01). Introspect tests a view on complete groups of keys read from both tables, and a keyed lookup over the combinations of the reference key's sources: row columns whose values lie in the key column's domain, and constants for key columns with at most 20 values.

Files without declared keys. The Parquet release declares no keys and names two columns differently from the SQL scripts (OrderDate is DT in orders, LineNumber is RowNumber in orderrows). A fact's pair (link k, integer column c) is taken as its line key when c is distinct within k and numbers the rows from 0 or 1 without gaps on complete groups of k (at least 99.9% of 30 or more groups with two or more rows; chance does not number thousands of rows this way). Proposition 61(a) needs only that V's columns equal columns of B on the joined rows, not that they share names: the key columns are paired by name with one left on each side paired with the other, and a column of V equal on 99.9% of the joined rows to exactly one unused column of B is read from it (`base_column`).

Precedence among drafted structures. A ratio $y = x\,f$ is not drafted when x is computed from y through derive rules or ratios already chosen (AdventureWorks: DealerPrice = 0.6 ListPrice with ListPrice = StandardCost $\times f$ and StandardCost = DealerPrice $\times f'$ was a cycle), and a column given by a derive rule is not a column of an implied dimension (ReorderPoint = 0.75 SafetyStockLevel keeps its formula, which holds for any level). The plan rejects any spec whose derived columns, ratio and offset columns included, depend on each other in a cycle.

12.3 Rules across tables

Proposition 62 (Conditioning links on rules: target and bias)**.** *Let an entity's links L be drawn given segments and attributes A from $Q(L \mid z)$ and let $\mathcal{R}$ be rules over (L, A) and the parents' columns. The independence Metropolis–Hastings kernel of Section 11 targets $Q(L \mid z, A, \mathcal{R})$ and leaves the law of A unchanged. (a) If $\mathcal{R}$ makes a class of a link a function of A (the online store iff the delivery gap is positive), the synthetic share of that class equals the synthetic share of the attribute event, so it has the attribute model's accuracy and no conditioning bias. (b) In general,*

if the fitted joint law Q of (A, L) is within total variation δ of the source law P and $P(\mathcal{R}) = 1$, then $\mathrm{TV}(Q(\cdot \mid \mathcal{R}), P) \le 2\delta$: conditioning on rules the source satisfies at most doubles the model's distance from the source.

Proof. The kernel moves links only; its acceptance ratio is 1 on valid states, so it is a Metropolis–Hastings kernel for the restricted law, and A is never resampled. (a) On valid states $\mathbf{1}[L \in C] = g(A)$. (b) $Q(\mathcal{R}) \ge P(\mathcal{R}) - \delta = 1 - \delta$; $\mathrm{TV}(Q(\cdot \mid \mathcal{R}), Q) = 1 - Q(\mathcal{R}) \le \delta$; the triangle inequality gives 2δ. □

Introspect drafts two kinds of cross-table rules, each tested against its permutation chance (Proposition 42): a date of the row inside a parent's date window (`days(StoreKey.OpenDate) <= days(OrderDate)`; a null bound is open), and an equality between columns of two parents, or of the row, with one exception member of a copied parent (`CustomerKey.Country == StoreKey.CountryCode or StoreKey == 999999`), with its converse when the exception never satisfies the equality. Functions `days` and `add_days` compare and shift dates whatever their storage (ISO text, datetimes, day numbers).

Rules within a row. Introspect also drafts order rules between two numeric or date columns of the same row (`Cost < Price`, `days(StartDT) <= days(EndDT)`), with the same test against the share of permuted pairs that satisfy the order. Ratio, offset and derived columns are skipped, since their order already follows from their construction. These rules read no link and are enforced by redrawing a row's attributes from its own conditional until they hold, so Proposition 62(b) bounds their cost; the first v0.6 run (before these rules) left 1.5% of Contoso's products priced below cost.

Proposition 63 (Calibrated exact draws of links under rules)**.** *Let an entity have no unique key over its links, so that given segments its rows' links are independent with laws Q_i, and let $\mathcal{R}$ be its key rules. Let $c(L)$ be the row's cell (the joint of the categorical columns, at most 50 levels each, that the rules read through the links: the customer's country × the store's country) and $t_j(L)$ the segment of link j's parent. For weights $w > 0$ on cells and $u_j > 0$ on parent segments let*

$$\pi_i(L) \propto Q_i(L)\, \mathbf{1}[\mathcal{R}]\, w(c(L)) \prod_j u_j(t_j(L)).$$

(i) $\pi_i(\cdot \mid c, t) = Q_i(\cdot \mid c, t, \mathcal{R})$: within a cell and parent segments the law is the model restricted to the rules. (ii) A row drawn as follows has law π_i: a cell c with probability $\propto w(c) \prod_j \sum_t \phi_{j,z_i t} u_j(t) W_j(t, c_j)$; for each link a segment $t \propto \phi_{j,z_i t} u_j(t) W_j(t, c_j)$ and a parent of t with value c_j by the link's member law; the row is kept if $\mathcal{R}$ holds and drawn again otherwise. (iii) The weights solving $\pi(c) = P(c)$ and $\pi(t_j) = \bar{Q}(t_j)$ (the cell margin of the source, and the parent-segment margins the fitted link model gives before conditioning) give the I-projection of the restricted law onto those margins (Csiszár 1975); iterative proportional fitting of w and u converges to it when the margins are attainable.

Proof. (i) The tilt is constant on each (c, t). (ii) The draw before the check has probability $\propto w(c) \prod_j \phi_{j,z_i t_j} u_j(t_j)\, \omega_{j,t_j}(L_j) = w(c) \prod_j u_j(t_j)\, Q_i(L)$ on the links with cell c, where $\omega_{j,t}$ is the member law of segment t ($W_j(t, c_j)$ cancels between the two stages); rejection on $\mathcal{R}$ restricts it. (iii) The I-projection onto linear (margin) constraints is an exponential tilt by functions of the constrained variables, i.e. of this form (Csiszár 1975, Theorem 3.1); IPF is its cyclic coordinate ascent. □

Without calibration Contoso's orders came 66% from US customers against 51% in the source, because the store-country rule rejected more proposals for the countries with fewer stores (K10); with the calibrated draws the cell margin is within 0.01 in total variation. The engine calibrates w only and keeps $u_j = 1$: adding the parent-segment margins $\bar{Q}(t_j)$ as a second set of constraints helped on S11 but oscillated on Contoso, where the rules make them unattainable together with the cells (Proposition 63(iii) needs attainable margins); their distance before and after is logged.

What remains is a shift in who buys: conditioning on a rule that the degree model already reflects (an order date inside the customer's active window; the source's counts per customer already grow with the window) concentrates the links on the customers that satisfy it most often. On S11 the share of customers with an order is 0.53–0.55 against 0.57 (0.58–0.60 without the rules); on Contoso 0.48 against 0.50, with 1.86 orders per buying customer against 1.79. Separating a customer's exposure (the orders it is eligible for) from its propensity in the degree model would remove it; it is listed as open. A row whose own segment mixture gives the rules no chance (one Contoso order whose customer segment never used the online store) is drawn after 100 attempts from the link's overall segment mixture $\sum_k \hat{\pi}_k \phi_{j,k\cdot}$, which stays inside the fitted model's support; rows still invalid after 100 more go to the Metropolis–Hastings kernel of Proposition 62, which also serves entities with unique keys over their links. A cell coordinate that a stratified link fixes cannot be calibrated: its margin is that of the strata. A rule that reads only stratified links and the row's attributes (the order date before the store's closing date, with the store fixed by the stratum) cannot be repaired by moving links at all; it is a rule on the attributes given the links and is enforced by redrawing the attributes, together with the link rules that read the same attributes (Proposition 62). On S11 five orders were left violating such rules after the repair rounds once the store became a stratifier.

Proposition 64 (Keys in date order)**.** *Synthetic surrogate keys are labels the model never reads. Assigning the j-th smallest key to the row with the j-th smallest value of a column c (ties in row order) is a permutation of labels: the joint law of all other values is unchanged and keys sorted by number are sorted by c on every row, as in the source (Contoso order numbers, K05).*

Proof. Children reference rows, not key values. The numbering is applied when the rows are drawn and again after the attribute redraws of the rules (which can move the ordering column), with the same permutation applied to every link that already points at the table. □

Proposition 65 (Conditional inclusion on a fact's anchor)**.** *Let a fact's anchor rows in segment k have i.i.d. row counts with law P_k ($P_k(0) = P0_k$; for an occupancy fact the Bernoulli activity with $P0_k = 1 - \pi_k$), and let the source satisfy a conditional inclusion dependency (Bravo, Fan and Ma 2007): anchor rows at an excluded level of a column (employees with `SalesPersonFlag = 0`) have no fact rows, a share e_k of segment k. Then the eligible rows' counts follow*

$$q_k(0) = \frac{P0_k - e_k}{1 - e_k}, \qquad q_k(c) = \frac{P_k(c)}{1 - e_k} \ (c > 0),$$

which keeps the segment's law of counts over all its rows. A draw from P_k that is zero and kept with probability $r_k = 1 - e_k/P0_k$, and drawn again otherwise, has law q_k. For the Bernoulli law this is activity $\min(1, \pi_k/(1 - e_k))$*. When $e_k > P0_k$ no such law exists, $r_k = 0$, and the shortfall is logged.*

Proof. $P_k = e_k\delta_0 + (1 - e_k)q_k$ gives q_k. Under the rejection, $\Pr(\text{output} = 0) \propto P0_k r_k$ and $\Pr(\text{output} = c) \propto P_k(c)$ for $c > 0$; with $r_k = 1 - e_k/P0_k$ the normaliser is $1 - e_k$ and the ratios are those of q_k. The DP count law is i.i.d. given its Bayesian-bootstrap weights, so these are drawn once and $P0_k$ is computed from them. □

The engine fits these rules on every link with at least 10 expected references to the excluded level and none observed, and masks the excluded rows when the link chooses a parent, but it skipped a fact's anchor, whose rows are not chosen but given counts. On AdventureWorks the employees' segments usually separate the 17 sales staff, so quotas went to sales staff by chance; after one segment merge 42 of 183 synthetic quotas named non-sales employees. The rule is now also fitted for the anchor. A Poisson anchor without the DP law has its row weights masked and rescaled within the segment instead. Check C52 forces a single employee segment: without the rule 165 quotas and 75 bonuses reference non-sales staff, and with it none do.

The partner of an occupancy fact. The same rule on the partner side (order lines never name a product without a price) was applied by masking the partner's row weights, which the Poisson block model uses but the occupancy model does not: it draws each (anchor, partner) pair of block (k, l) with probability p^*_{kl} over all partner rows of segment l. One AdventureWorks reseller line in 60,855 named a product without a price, and so had no unit price. The excluded partner rows had no pairs in the source, so the block's pairs belong to its eligible rows: they are drawn over the E^*_l eligible synthetic rows only, with $p^*_{kl} = \min(1, p_{kl} m_l / E^*_l)$. This is the scale rule with E^*_l in place of m^*_l, and keeps the expected pairs per anchor, $p_{kl} m_l$. Check C56 forces one product segment: before the change 2,251 of about 7,200 lines named never-sold products, and after it none do, with lines per order within 1%.

Mandatory participation under the DP count model. When the counts per parent are modelled by the Dirichlet process (Section 11), new atoms come from the Poisson base measure, which puts mass $e^{-\lambda}$ on 0 even when participation is mandatory (every order has a line); one synthetic Contoso order had no lines. With mandatory participation the base measure is $1 + \text{Poisson}(\lambda - 1)$, the same shift the parametric model uses: the support is that of the source and the mean is unchanged. Both the unconditional draws and the draws given a segment's total use it (check C54).

Self-anchored dimensions. A dimension of fewer than 200 levels with no categorical label of at most 50 levels to anchor it (AdventureWorks resellers: the number of employees determines annual sales) was drafted as a link. The code's dependence on the other columns then rested on the segments alone, and business type with number of employees fell from 0.91 to 0.06. When the marginal model keeps the code exactly (a category, or an integer with at most 30 values) the code now anchors itself: it stays a modelled attribute, and the columns it determines are keyed lookups into the copied dimension by its value. The dependency then holds on every row, and the copula keeps the dependence. Otherwise no dimension is drafted. On AdventureWorks the prospective buyers' state code (38 values, which determines one column) is self-anchored, and the resellers' 76 employee counts are left to the copula (business type with number of employees 0.84 against 0.91). Check C55 tests both properties.

12.4 Cost of stratification

A Gibbs sweep scores every row of an entity against every segment: $O(nK)$ per attribute and link. Strata (rows of copied parents) nest segments, $K = \sum_s \text{cap}_s$, so a 555-level geography with 105,000 customers gave $K = 1{,}757$ and 21 s per sweep (0.6 s before). Stratifiers are now dropped, largest first, while nK exceeds `segments.stratify_max_cells` ($5 \cdot 10^7$); a dropped stratifier's link is modelled by the degree-corrected block model like any other link (Proposition 46), and the decision is logged. AdventureWorks (18,484× about 700) stays stratified. When stratifiers must be dropped, implied dimensions go first, whatever their size: their anchor keeps them as modelled attributes, while a real copied parent's link has no other carrier. In the first v0.6 AdventureWorks run the customers' education and occupation dimensions (5 levels each) were kept and the geography dropped, and the share of synthetic customer combinations not seen in the source rose to 0.12; excluding implied dimensions altogether fixed that but took the subcategory strata from Contoso's products (Weight with SubCategoryName 0.85 in the source, 0.52 synthetic).

13 Bayesian synthesis and the whole data model (v0.7)

Version 0.7 makes every synthetic dataset a draw from the posterior predictive distribution, closes the open items of the Contoso and AdventureWorks reports (T5, T11, T14, the participation share and the quota segment), and adds the warehouse structures that the four databases did not all contain but that are standard practice (Kimball and Ross 2013): type 2 slowly changing dimensions, unknown members and bridge tables. Each structure is detected by `introspect`,

written to the specification and can also be declared by the user (`docs/spec-reference.md`). Checks C57–C68 test this section on fake data with known truth.

13.1 Posterior predictive synthesis

Proposition 66 (One parameter draw per dataset)**.** *Let θ collect every parameter block of the model (segment rates, Dirichlet weights, copula correlation, null shares, the shares below) with posterior $p(\theta \mid D)$. Dataset i draws $\theta^{(i)} \sim p(\theta \mid D)$ once from its own parameter stream and then every row from $p(\cdot \mid \theta^{(i)})$ with a separate data stream. (i) $D^{*(i)}$ is an exact draw from the posterior predictive $\int p(D^* \mid \theta)\, p(\theta \mid D)\, d\theta$ whenever each block's draw is exact (conjugate blocks) and approximately so for the MCMC blocks. (ii) For a statistic q, $\mathrm{Var}(q(D^*)) = \mathbb{E}[\mathrm{Var}(q \mid \theta)] + \mathrm{Var}(\mathbb{E}[q \mid \theta])$; drawing θ afresh per row instead removes the second term asymptotically, so the between-dataset variance b_m of the combining rules for fully synthetic data, $T_m = (1 + 1/m)b_m - \bar{u}_m$ (Raghunathan, Reiter and Rubin 2003), would be too small and intervals would under-cover. (iii) Datasets $i \neq j$ are independent given D.*

Proof. (i) Composition sampling. (ii) Law of total variance; with per-row draws the rows are i.i.d. from the predictive $\int p(x \mid \theta)p(\theta \mid D)d\theta$, so $\mathrm{Var}(\mathbb{E}[q \mid \cdot])$ of a mean of n rows is $O(1/n)$ of the per-row variance instead of the posterior variance of $\mathbb{E}[q \mid \theta]$. (iii) The streams are spawned from independent children of one `SeedSequence`. □

`posterior.mode: mean` replaces each draw by its posterior mean (each spike-and-slab indicator by its posterior median): the plug-in synthesis of Reiter and Kinney (2012), stable from run to run and meant for test data. Kept as point values on purpose: deterministic identities (lookups, derives, aggregates, ratios, offsets), copied tables, storage precision, privacy thresholds, user-set sizes, and model-structure choices that a Bayes factor settles (segment counts after merging, structural zeros above the prior threshold, count laws, the frailty switch). Averaging over structure would change the table layouts between datasets. Two blocks keep an empirical Bayes estimate: the Gamma frailty shape α (a moment estimate from the dispersion index) and the copula sampler's hyper-parameters; Proposition 70 corrects the one consequence of $\hat{\alpha}$ that the checks exposed (the share of parents without children). *Check C57*: a dataset is reproducible from its index, datasets differ by the posterior spread, and `mean` mode is stable.

13.2 Rules that hold on most rows

Proposition 67 (Soft rules)**.** *For a rule $y = f(x)$ that holds on h of n source rows, draw $\lambda \sim \mathrm{Beta}(1+h, 1+n-h)$ and offset weights $w \sim \mathrm{Dir}(c+\frac{1}{2})$ over the observed offsets $d_j = y - f(x)$ (counts c_j) once per dataset; set $y = f(x)$ with probability λ, otherwise $y = f(x) + d_J$ with $J \sim w$ (or the modelled value of y when there are no offsets or more than 50). (i) The expected share of synthetic rows on which the rule holds is $\mathbb{E}\lambda = (1+h)/(2+n)$, and its posterior predictive spread is that of the Beta-binomial. (ii) Every synthetic violation is an observed offset, so a rule that fails by one year (an age before the birthday) fails by one year. (iii) As $n \to \infty$ the share converges to h/n.*

Proof. Beta–binomial and Dirichlet–multinomial conjugacy; (ii) by construction. □

`introspect` finds such rules for an age or a number of whole years between two dates: the reference date is swept over the interval that each row's value implies and kept when the rule holds on at least 80% of rows, far above the share that permuting y gives. Contoso's customer age holds on 90.5% of source rows; v0.6 gave 42%, v0.7 gives 90.4% (T14). *Check C58.*

13.3 Values shared by a group

Proposition 68 (Group effects)**.** *Let rows be grouped by a key g (columns of the row or of its parents, such as the product category and the order date), and let each row draw $U_r \sim \mathrm{Unif}(0, 1)$*

once. The group's value is $u_g = y_{r^*}$ with $r^* = \arg\min_{r \in g} U_r$, the value that the row model drew for that row, and every row of g takes u_g. (i) The rule "one value per group" holds on every synthetic group. (ii) r^* is uniform on the rows of g, so u_g has the law $\frac{1}{|g|}\sum_{r \in g} p(y \mid x_r, \theta)$: a shared random effect whose distribution is the row model's averaged over the group (Hu, Reiter and Wang 2018 use the same construction for households). (iii) Redrawing some rows of g (key rules) does not change r^* unless r^* itself is redrawn, so the recomputation after redraws is stable. (iv) For a ratio column $y = \text{round}(xf)$ the effect is placed on the factor f, so rows of a group share the factor and keep their own base x.

Proof. (i) By assignment. (ii) The minimum of i.i.d. continuous uniforms is attained at a uniformly random index. (iii) The priorities are stored with the rows. (iv) Proposition 59(b). □

Detection. With s the share of groups of two or more rows that hold one value (the g_3-type measure; Parciak et al. 2024) and p_0 its permutation mean, a grouping is proposed when $s \geq 0.999$ and the test of Proposition 55 passes; among groupings that pass, the coarsest is kept (fewest groups). Contoso's price factors are one per category and day on 100% of source groups; v0.6 drew them per row (30% of groups with one factor), v0.7 gives 100% (T11). *Check C59.*

13.4 Two links that agree more often than chance

Proposition 69 (Link affinity)**.** *Let a row have links a and b whose parents carry an attribute v (the customer's area and the store's area). Under the link model Q the share of rows with $v(a) = v(b)$ is π_Q; in the source it is h/n. Model*

$$P(b \mid row) = (1 - \rho)\, Q(b \mid row) + \rho\, Q(b \mid row,\, v(b) = v(a)),$$

the diagonal-inflated quasi-independence table of Goodman (1968), with ρ set so that the share of matching rows equals $s \sim \text{Beta}(1 + h, 1 + n - h)$, drawn per dataset. The engine realises it by moving $\lceil sn \rceil - m$ non-matching rows (m already match) to a parent with value $v(a)$, chosen with probability proportional to the parent's degree $+\frac{1}{2}$, keeping a move only when every key rule of the row still holds. (i) Whenever enough non-matching rows have a matching parent, the synthetic matching share is $\text{round}(sn)/n$ exactly, with posterior predictive spread $\text{Beta}(1 + h, 1 + n - h)$. (ii) Moved rows keep their other columns; within the matching parents the choice follows the link's popularity (a Dirichlet$(\deg + \frac{1}{2})$ posterior mean). (iii) Rows are only moved toward matching parents, so the share of rows matching on a coarser attribute (country when v is the area) does not decrease.

Proof. (i) Each accepted move turns one non-matching row into a matching one and no move does the reverse. (ii) By construction. (iii) $v(a) = v(b)$ implies equality of every function of v. □

Detection. For each pair of links and each pair of attributes of their parents, the lift s/π_0 over the analytic permutation mean $\pi_0 = \sum_v p_a(v) p_b(v)$ (the expected equality share when one column is permuted) must be at least 2 with the binomial test of Proposition 42, for attributes with the same name or sharing at least half their values; the finest such attribute (most values) is kept, and the link to the smaller parent is the one moved. When the equality holds on at least 99.9% of rows it is a rule $a.v = b.v$ (a `check`), not an affinity. Contoso: customer and physical store in the same geography area on 12.6% of source lines; v0.6 gave 1.6%, v0.7 gives 12.1% (T5). *Check C60.*

13.5 Parents without children and segments that mix eligible rows

Proposition 70 (Participation hurdle)**.** *For a link from an entity to a parent with segments t, let Z_t of the N_t source parents of segment t have no child; draw $s_{0t} \sim \text{Beta}(1 + Z_t, 1 + N_t - Z_t)$ per dataset (Mullahy 1986). After the links are drawn, in each segment with more empty eligible parents than $\text{round}(s_{0t} N'_t)$, children are moved from parents of the same segment with two or more*

children (chosen with probability $\propto$ degree -1) to empty eligible parents, each move kept only when the child's key rules hold. Moves stay within a segment because the child's other columns were drawn given its parent's segment (what a customer of that segment buys); codes implied inside a table (Proposition 57) are drawn with their anchor and are not moved. (i) The number of children is unchanged, so the mean degree is unchanged. (ii) No move creates an empty parent. (iii) When enough movable children exist, the share of empty parents in segment t is $\mathrm{round}(s_{0t}N'_t)/N'_t$.

Proof. (i) Moves change only the parent. (ii) The source parent keeps at least one child. (iii) Each move fills one empty parent. □

A Gamma frailty with shape α and rate λ gives a zero share $(1+\lambda/\alpha)^{-\alpha}$; with the moment estimate of α this missed Contoso's share of customers with an order (0.48 against 0.50). The hurdle fixes the zero share without changing the number of children. *Check C61.*

Proposition 71 (Cannot-link strata)**.** *Let a categorical column c of an entity have a level ℓ whose n_ℓ rows are never referenced by a child relation that references D of the n rows, with $Dn_\ell/n \geq 10$. Under independence of reference and level, the probability of no reference is at most $\exp(-Dn_\ell/n) \leq e^{-10}$, so ℓ is a structural zero. Stratifying the entity's segments by the indicator $\mathbf{1}[c=\ell]$ (a cannot-link constraint, Shental et al. 2003) gives segments that never mix the two kinds of rows, and the child's link model gives the excluded segments rate zero, so no synthetic child references an excluded row.*

Proof. Sampling D referenced rows without replacement: $P(\text{none in } \ell) = \binom{n-n_\ell}{D}/\binom{n}{D} \leq (1-n_\ell/n)^D \leq e^{-Dn_\ell/n}$. Within strata the segment rates are fitted separately, and a stratum with no source children has posterior rate mass only from the prior, which the structural-zero test of Section 9 sets to zero. □

AdventureWorks: sales quotas exist only for sales staff; the employee segment that mixed sales staff with others made the quota count move between runs (138–173 against 163). Small tables (fewer than two segments' worth of rows) keep the earlier attribute strata and get exact stratum counts. *Check C62.*

13.6 Warehouse structures: slowly changing dimensions, unknown members, bridges

Proposition 72 (SCD type 2)**.** *A type 2 dimension stores versions of a natural key k with validity intervals $[f_v, t_v]$, the latest open (t = null) or closed. Fit: versions per key $V \sim \mathrm{Dir}(c+\frac{1}{2})$ over $1..M$; the share of keys whose latest version is open $\sim \mathrm{Beta}(1+o, 1+n_k-o)$ (used when there is no valid-to column); the gap $t_v - f_{v+1}$ in days $\sim \mathrm{Dir}(c+\frac{1}{2})$ over the observed gaps; stable columns (equal across a key's versions on $\geq 99\%$ of keys with two or more versions). Synthesis groups the synthetic rows into keys with sizes drawn from V (rows of one segment together). A row whose modelled t is null (the row model draws the null with the row's segment, under the null-share posterior) becomes the latest version of its key: open rows are spread one per key by swapping with closed rows of keys that have none (same segment first; keys never span two strata, so a stratifying column stays constant per key), and within a key the sorted start dates go to the closed rows first and to the open row last. The key and the stable columns are copied from each key's first row and $t_v = f_{v+1} + \text{gap}$ for every earlier version. (i) Every key has at most one open version and its versions do not overlap when the gaps are non-positive (0 or -1 day, as in every clean history). (ii) The number of keys is $n/\mathbb{E}V$ in expectation and the version-count law is the posterior predictive. (iii) Stable columns are constant within each key. (iv) The current flag, when declared, equals its value exactly on the open versions. (v) When the open rows can be spread (at most one per key), every row keeps the open or closed status the row model gave it, so children linked to a row by the segment model keep the source's share of links to open versions.*

Proof. (i) Only the latest version may be left open, and $t_v = f_{v+1} + \text{gap} \le f_{v+1}$ with $f_v < f_{v+1}$. (ii) Renewal argument: groups are drawn until n rows are covered. (iii), (iv) By assignment after grouping. (v) Swaps move rows between keys, not statuses; only the extra open rows of a key that could not be swapped are closed. □

Detection needs a valid-from and a valid-to date, a candidate natural key repeated on some rows, distinct start dates within a key and at most one open version per key (the latest), on at least 99.9% of keys, with at least 5 keys holding two or more versions. AdventureWorks: `DimProduct` (504 product codes, 77 with two or more versions) and `DimEmployee` (290 national IDs, 5 with two or more); `DimCustomer` is correctly not one. Facts keep pointing at surrogate keys, which is how a type 2 dimension is joined. *Check C63.*

Proposition 73 (Unknown members)**.** *Let a dimension contain placeholder rows U (key -1, 0 or a label such as "Unknown") that r child rows reference while m child rows have a null key. The placeholders are removed before fitting and copied to the output, and their references become nulls. After synthesis each null reference is set to a placeholder with probability $q \sim \text{Beta}(1+r, 1+m)$ (the member chosen by $\text{Dir}(c + \frac{1}{2})$ over the placeholders). (i) Placeholders never enter the dimension's segments or the facts' rates, so the one "Unknown" row cannot become a segment of its own or pull the attribute model toward its nulls. (ii) The expected share of references to placeholders among null-or-placeholder references is $(1 + r)/(2 + r + m)$. (iii) A fact drops rows whose dimension key is null from its block model; for facts the placeholder rows are therefore restored by cloning $\text{round}(q' n^*)$ synthetic rows chosen uniformly ($q' \sim \text{Beta}(1 + r, 1 + n - r)$) and pointing them at a placeholder. This assumes the source rows that reference a placeholder are missing at random with respect to the other columns; the user declares otherwise by modelling the placeholder as a member (not listing it in `unknown_members`).* Check C64.

Proof. (ii) Beta–binomial. (iii) Under missing at random the rows with an unknown member have the same conditional law as the other rows, which a uniform clone reproduces in expectation. □

Proposition 74 (Bridge weights)**.** *In a bridge table (accounts of a customer, diagnoses of a stay) the weights w_i of a group sum to a total T (1 by default). The modelled weights are normalised, $w_i \leftarrow T w_i / \sum_{j \in g} w_j$, rounded to d decimals and the rounding remainder added to the group's largest weight. (i) Each group's weights sum to T to the stored precision. (ii) Ratios of weights within a group are those of the model. (iii) If the modelled weights are independent $\text{Gamma}(a_i, 1)$ the normalised weights are $\text{Dir}(a)$ (the standard construction), so a group's allocation has the Dirichlet law with the model's shapes.*

Proof. (i) After rounding, the remainder is a multiple of 10^{-d} smaller than the largest weight. (ii) Scaling. (iii) The Gamma representation of the Dirichlet. □

Detection: a fact whose float column sums to the same value (1 or 100) on at least 99.9% of the groups (with two or more rows) of a foreign key. *Check C65.*

13.7 Identifying the whole model

Proposition 75 (Equality through two links)**.** *Let a and b be columns (possibly through links) of the n rows of a table, with level shares $p_a(v)$ and $p_b(v)$. Under a uniform random permutation of b the expected share of rows with $a = b$ is $\pi_0 = \sum_v p_a(v)\, p_b(v)$. A rule $a = b$ is reported when the observed share is at least 0.999, above π_0, and passes the test of Proposition 55.*

Proof. $\mathbb{E} \frac{1}{n} \sum_i \mathbf{1}[a_i = b_{\pi(i)}] = \frac{1}{n} \sum_i P(b_{\pi(i)} = a_i) = \frac{1}{n} \sum_i n_b(a_i)/n = \sum_v p_a(v) p_b(v)$. □

The closed form replaces a Monte Carlo estimate, and constant groups are screened in $O(n \log n)$ by sorting, which made `introspect` on AdventureWorks (27 tables) feasible with the new detections. `relsynth model` writes the identified model with Kimball's vocabulary: facts (transaction, periodic

snapshot, accumulating snapshot, factless), dimensions (conformed, role-playing, degenerate, outrigger, SCD type 2, bridge), hierarchies with their levels, rules, and the join graph as a Mermaid entity-relationship diagram.

13.8 The specification

Every key the specification accepts has a type, a default and a description in one schema table (`relsynth/schema.py`); unknown keys are errors with the closest valid key as a hint (*check C67*), and the key reference in `docs/spec-reference.md` is generated from that table. Column laws can be stated instead of learned (`distribution`: fixed, categorical probabilities, uniform, normal, lognormal, Poisson, date range, sequence, empirical); a stated law replaces the learned one and is drawn independently of the other columns (once per row, so redraws of other columns keep it); `empirical` resamples the source values with Bayesian-bootstrap weights. Structures the data cannot reveal are listed as user guidance (Section 4 of the reference); *check C66* runs its recipes. *Check C68* runs the reference's annotated example, which uses every v0.7 option at once.

14 Identification from the data, exact speed-ups and the exported model (v0.8)

Version 0.8 identifies keys and links from the data when the catalog declares none, makes the engine faster without changing a single output bit, corrects two run-length rules that the Contoso data showed to be wrong, and exports the fitted model as a self-contained mathematical object (`relsynth export-model`). Checks C69–C74 test this section.

14.1 Keys and links from the data

No schema template, naming list or known database is used: the evidence is the values, and optionally the column and table names weighted by how rare each word is in the database at hand (inverse token frequency, so words every table carries count for nothing; `-evidence data` switches names off). Primary keys are unique column combinations scored as in holistic key detection (Jiang and Naumann 2020). A link is a unary inclusion dependency $c \subseteq K$ with inclusion share $\sigma \geq 0.95$; it counts as evidence only when it is unlikely by chance.

Proposition 76 (Inclusion by chance)**.** *Let the $v \geq 3$ distinct integer values of c that lie in K have extremes $a < b$, let $w = b - a + 1$ and let $k = |K \cap [a, b]|$. Under the null that, given the extremes, the other $v - 2$ distinct values are a uniform draw without replacement from the $w - 2$ interior integers, the probability that all of them fall in K is*

$$P = \binom{k-2}{v-2} \Big/ \binom{w-2}{v-2} \;\leq\; d^{\,v-2}, \qquad d = k/w.$$

A key that fills its range ($d = 1$, a surrogate $1..N$) gives $P = 1$: inclusion in it is no evidence at all, and the decision falls to the other features (randomness, coverage, names) or to a question.

Proof. The extremes are keys because every value counted lies in K. A uniform $(v-2)$-subset of the $w - 2$ interior points lies in the $k - 2$ interior keys with the hypergeometric probability shown. Each factor $(k - 2 - i)/(w - 2 - i) \leq k/w$ for $k \leq w$, which gives the bound. □

Proposition 77 (Randomness bound)**.** *Let $u_1, \ldots, u_m \in [0, 1]$ be the quantile positions of the child values in the sorted parent key and F_m their empirical distribution. Then $W_1(F_m, U(0,1)) = \int_0^1 |F_m(x) - x|\, dx \leq \frac{1}{2}$, with equality only for a point mass at 0 or 1, so $R = 1 - 2W_1 \in [0, 1]$. If the child values are a random sample of the key, $R \to 1$ as $m \to \infty$.*

Proof. For a point mass, $W_1(\delta_a, U) = \int_0^1 |a - x| dx = \frac{a^2 + (1-a)^2}{2} \le \frac{1}{2}$, with equality at $a \in \{0, 1\}$. Any F on $[0, 1]$ is a mixture $\int \delta_a \, d\mu(a)$, and $|F(x) - x| = |\int (\mathbf{1}[a \le x] - x) \, d\mu(a)| \le \int |\mathbf{1}[a \le x] - x| \, d\mu(a)$; integrating over x gives $W_1(F, U) \le \int W_1(\delta_a, U) \, d\mu \le \frac{1}{2}$. For a random sample, $W_1(F_m, U) \to 0$ almost surely (Glivenko–Cantelli on a bounded interval). The implementation integrates $|F_m(x) - x|$ exactly piece by piece. □

Proposition 78 (Chance of a forest)**.** *A table of n rows with $r \ge 1$ null pointers, whose other $m = n - r$ pointers are drawn independently and uniformly from the n keys (a self-pointer counting as a top), has pointers that form a forest with probability*

$$p(n, r) = \frac{r+1}{n}\Big(1 + \frac{1}{n}\Big)^{m-1} \;\le\; \frac{e\,(r+1)}{n}.$$

In v0.8 a forest was accepted as evidence of a self-reference when $p(n, r) \le 0.01$; otherwise the link needed other evidence (names, Proposition 76) or was asked. v0.9: *$p(n, r)$ is the chance under* uniform *pointers, which a code column (values* 1..3 *in a table of 2,000 rows, its few targets pointing at themselves) does not follow; the decision now uses the permutation test of Proposition 96, and $p(n, r)$ remains as reported context.*

Proof. A rooted forest on n labelled vertices with a given set of k roots is a map from the other $n - k$ vertices to all n vertices without cycles; there are $k\, n^{n-k-1}$ of them (Cayley's formula for forests). Let s of the m non-null pointers point to their own row; the other $m - s$ then form a forest rooted at the $r + s$ tops. Summing over s and dividing by the n^m maps,

$$p = \sum_{s=0}^{m} \binom{m}{s} \frac{(r+s)\, n^{m-s-1}}{n^m} = \frac{1}{n}\Big[r\Big(1 + \tfrac{1}{n}\Big)^m + \frac{m}{n}\Big(1 + \tfrac{1}{n}\Big)^{m-1}\Big] = \frac{r+1}{n}\Big(1 + \tfrac{1}{n}\Big)^{m-1},$$

using $\sum_s \binom{m}{s} x^s = (1+x)^m$ and $\sum_s s\binom{m}{s} x^s = mx(1+x)^{m-1}$ with $x = 1/n$, and $r(1 + \frac{1}{n}) + \frac{m}{n} = r + 1$. Finally $(1 + \frac{1}{n})^{m-1} \le e^{(m-1)/n} \le e$. □

The earlier documentation said that a random mapping "almost surely has a cycle"; that holds only without null pointers. With r tops the chance is about $e(r + 1)/n$: 0.51 for Chinook's 8 employees with one top, 0.018 for AdventureWorks' 296 employees with one top. From values alone both are now asked instead of being accepted on weak evidence; with names, AdventureWorks' is found (its key words recur across the database) and Chinook's is still asked, and the declared catalog settles both when it has them. *Check C71* verifies the three statistics against simulation and exact integrals; *check C70* hides every declared key and link of the retail database and finds 8 of 8 links with names and 0 false links in both modes, with a question for each link that values alone cannot establish.

A column whose name shares a word with at least half of this database's primary-key names (learnt from the keys found, such as *id* or *key*) and whose values all lie in another table's key is asked about when its values do not establish the parent; with `-evidence data` names are not read and only the value evidence above applies.

14.2 Exact speed-ups

Proposition 79 (Exact kernels)**.** *Each of the following replaces an expression by code that performs the same IEEE-754 operations on each element in the same order and consumes the same random numbers, so outputs and the state of every random stream are bit-identical: (a) the Gaussian log-likelihood $L_{ik} \mathrel{+}= a_{kj} - \frac{1}{2}(x_{ij} - \mu_{kj})^2 / s_{kj}^2$ computed column by column (compiled when Numba is installed); (b) count tables by one* `bincount` *instead of* `add.at` *(integer counts are exact in floating point below 2^{53}); (c) Dirichlet rows: NumPy draws* $\mathrm{Dir}(\alpha)$ *for* $\max_j \alpha_j \ge 0.1$ *(corrected in v0.9: the v0.8 text said* min*; below that NumPy switches to stick-breaking) as standard gammas g_j in order, multiplied by $1/\sum_j g_j$, which is reproduced in one call; (d) categorical draws*

`choice`(n, p) *as* `searchsorted` *of the normalised cumulative sum at one uniform per draw; (e) the Gumbel-max draw* $\arg\max_k(\ell_k - \log(-\log u_k))$ *in one buffer; (f) the occupancy likelihood of a row computed only from the terms that row uses (a present pair or the absence term), the others never being read; (g) frequency counts by value instead of by row (Kish effective size, canonical spellings).*

Proof. (a), (e) and (g) evaluate identical expressions on identical operands; (b) integer arithmetic; (c) and (d) follow NumPy's `Generator.dirichlet` and `Generator.choice` algorithms (uniforms drawn one per call equal those drawn as a vector, since `random(n)` consumes the stream in the same order); (f) the unread terms do not enter any output. □

Check C69 compares every kernel with the expression it replaces (and with a frozen copy of the previous occupancy code). On Contoso the fit is 2.5 times faster (Gibbs 3.7 times), sampling 2.1 times and evaluation 2.0 times, with every output file identical before the run-length changes below. Stan, Cython and C extensions were weighed: the remaining time is in vectorised NumPy calls already running compiled code, and the one loop that dominated (the Gaussian log-likelihood) is compiled by Numba when present; Stan is used for verification (Proposition 86), not for production sampling, where its general-purpose NUTS sampler would be slower than the conjugate updates.

14.3 Run-length control, corrected

The Contoso fit with v0.7 settings left 3 of 56 copula chains and the final Gibbs phase with split $\hat{R} > 1.1$. Two causes were found and fixed.

Proposition 80 (Fixed-width stopping for correlation chains)**.** *Let a chain of correlation matrices have kept draws* $C^{(1)}, \dots, C^{(T)}$, $T \geq 100$, *and let* $\hat{\sigma}_{ij}$ *be the batch-means standard error (batch size* $\lfloor\sqrt{T}\rfloor$*) of the mean of* $z_{ij} = \operatorname{atanh} C_{ij}$. *Stop when every entry has split* $\hat{R} \leq 1.1$ *or* $\hat{\sigma}_{ij} \leq \varepsilon = 0.01$, *and* $\log\det C$ *has split* $\hat{R} \leq 1.1$ *or standard error* $\leq 2p\varepsilon$. *(i) For a polynomially ergodic chain the batch-means estimator is consistent, so* $\bar{z}_{ij}$ *is within* 2ε *of its posterior mean with probability about 0.95 (Jones, Haran, Caffo and Neath 2006; Flegal, Haran and Jones 2008). (ii) An error* δ *on the* z *scale changes the conditional variance* $1 - \rho^2$ *by a relative amount of at most* $2|\rho|\delta \leq 2\delta$, *uniformly in* ρ, *which is why the tolerance is set on the Fisher* z *scale: near* $\rho = \pm 1$ *a fixed tolerance on* ρ *itself would be too loose.*

Proof. (i) Markov chain central limit theorem with a consistent variance estimator. (ii) $d(1 - \rho^2)/(1 - \rho^2) = -2\rho\, d\rho/(1 - \rho^2) = -2\rho\, dz$. □

Split $\hat{R}$ is scale-free: a nearly deterministic pair (Contoso's birthday and age, $\rho = -0.9989$, posterior s.d. 10^{-4}) drifts by 10^{-4} and shows $\hat{R} = 1.6$ although no synthetic value can reflect that drift; its standard error on the z scale is 0.003. The fixed-width branch is what accepts such a chain; it needs at least 100 kept draws so that the batch means are not computed on a chain shorter than its autocorrelation.

Proposition 81 (Rare columns per estimation sample)**.** *Proposition 29 applied to the sample a copula is fitted on: a column with fewer than* m *non-modal rows in a segment's sample (*$m =$`marginals.min_dependence_rows`*, default 5), which are also at most 1% of that sample, is kept independent of the other columns in that segment. Within the segment the joint law of that column with any other changes by at most 2% in total variation.*

Proof. Proposition 29 with the segment's sample as the population and $q \leq 0.01$. □

A column can be nearly constant within a segment although it is not in the whole table: Contoso's customer country took a second value on 3 of the 1,500 sampled rows of one segment, and those 3 rows sat at the extreme longitudes. The correlations of that column are then a function of 3 rows (quasi-separation in a probit model), the posterior is weakly identified and the chain wandered

($\hat{R} = 2.6$ after 3,000 iterations, chain means of one correlation from -0.12 to -0.65 over 4 seeds). The rule fixes it (3 rows are 0.2% of the sample). The 1% bound keeps the rule away from small segments, where a handful of rows is a sizeable share and independence would cost more than it saves; one segment of 28 still has a column (country, 16 and 2 rows away from the mode) whose correlations mix slowly, and the export reports it as a warning with the column pairs and the setting that removes it.

Proposition 82 (Burn-in after the merge)**.** *The merge step moves the segment chain to a new partition (a deterministic map of the state), after which the blocked Gibbs sampler (layer 1, Algorithm 3) needs a burn-in. The final phase is therefore monitored on the second half of its sweeps and extended in blocks while the split $\hat{R}$ of the attribute log-likelihood exceeds 1.1 (up to* `gibbs.max_final_iterations`*); the partition used is the last state, so the window changes only when the chain stops, not the kernel.*

Proof. Discarding an initial segment of a chain does not change its stationary law; the stopping rule reads only past states. □

On Contoso the log-likelihood falls for about 25 sweeps after the merge and is then flat: monitoring the whole final phase gave $\hat{R} = 1.23$ and a Geweke z of 8 even after 120 sweeps; monitoring its second half gives $\hat{R} = 1.00$ and $|z| < 2$ after 60 sweeps.

14.4 The exported model

`relsynth export-model spec.yaml` writes `model.json` (structure, posterior mean and M posterior draws of every parameter block, strict JSON), `model.md` and `model.pdf` (the generative model with this database's numbers and the verification table), `replicate.py` with `relsynth_expr.py` (a standalone sampler needing NumPy, pandas and SciPy) and `run_model.py` (the engine's own calls).

Proposition 83 (Exported draw m is the parameter of dataset m)**.** *Draw m of* `model.json` *is computed by the engine's parameter draw with the parameter stream of dataset m (Proposition 66), so it equals the $\theta^{(m)}$ from which* `relsynth run` *generates dataset m;* `run_model.py` *performs the calls of* `relsynth run` *and reproduces its output byte for byte.*

Proof. Same function, same seed sequence child, same order of draws; JSON writes floats by their shortest round-trip representation, which parses back to the same double. □

Checks C72 and C73: draws equal the engine's parameters for every dataset index tested; 18 of 18 files of two datasets are identical.

Proposition 84 (Verification conditions)**.** *Before the model is written, each parameter set (the mean and every draw) is checked against the conditions under which the generative process is defined and the propositions apply: (i) every probability vector on the simplex, every probability in $[0, 1]$, every rate non-negative and every quantile function non-decreasing (proper conjugate posteriors, Propositions 5, 17, 19, 2); (ii) every copula correlation matrix symmetric with unit diagonal and positive semi-definite, the condition of Sklar's theorem for a Gaussian copula (Propositions 1, 3); (iii) every block with a positive rate has source partner rows, the condition of the scale rule (Proposition 7); (iv) hierarchies drawn from level counts, so their size is finite whatever the spectral radius of the offspring matrix (Propositions 38, 54); (v) the Markov chains stationary by Propositions 82 and 80. A failure of (i)–(iv) is reported at the top of* `model.md` *and makes* `export-model` *exit with an error status; (v) is reported as a warning, because the draws are still states of the right chains.*

Proof. Each condition is a hypothesis of the proposition cited. □

Proposition 85 (The replica samples the core generative process)**.** *Given θ,* `replicate.py` *draws: entity segment sizes by largest remainder of $n\pi$; latent segments; attributes from the segment's Gaussian copula and marginals (given parent attributes, from the conditional normal with the given latent values uniform on their bins); hierarchies from the level-by-segment counts; links by segment and degree weight or DP counts; fact rows by the Poisson block or occupancy model; time columns from the anchor segment's period profile; then sequences, offset keys, lookups, derived columns and aggregates with the engine's own expression evaluator. Each step draws from the same conditional law as the engine, so the replica's output has the engine's law for the parts it covers; it is not bit-identical (different random streams). The parts it does not cover are listed in* `replica_scope` *(rules imposed by rejection or moves, slowly changing versions, affinity and participation moves, group effects, text and identifier models, nested facts and virtual dimensions).*

Proof. Step by step, the laws are those of Propositions 1, 3, 5, 54, 11 and 16 given θ. □

Check C73: no orphan keys and entity sizes exact on S9, S10 and S11; replica scorecards 104/105, 139/143 and 85/91 (the misses are the engine-only parts in the scope list).

Proposition 86 (Independent check with Stan)**.** *For the rate, occupancy, category, segment-attribute and copula blocks, the engine's posterior draws and those of a Stan program written in textbook form (prior and likelihood only) agree in mean within four Monte Carlo standard errors and in standard deviation within 20%.*

Proof. By computation (`relsynth.stancheck`); the two implementations share no code. □

Check C74: 11 of 11 parameters agree (the check is skipped when Stan is not installed).

15 Review: pooling, exact conditioning and calibrated limits (v0.9)

Version 0.9 is a review of the whole engine. Its common thread is a pattern found in the issues of earlier runs: *many units, each with its own fixed prior, threshold or test, and no pooling or multiplicity control*, give a steady trickle of small errors that no single unit shows. Examples are structural blocks drawn free in complete relations (bi B7), small segments given empty parents (AdventureWorks customers without a sale), a maximum over 48 cells compared with a fixed $z = 3$ (Contoso), a forest test whose null does not fit code columns, and per-check limits whose family produces false failures. The propositions below give the corrected models, samplers and limits. Checks `v09_core.py` (14), `v09_eval.py` (14), `v09_ident.py` (14) and `v09_io.py` (28), run as C75–C78, test this section.

15.1 Structural blocks pooled across a relation

Proposition 87 (Learned structural weights)**.** *Let the rows k of a relation fall in groups g (rows that share their structure; an ungrouped row is a group of its own), and let block (k, l) have S_{kl} successes in T_{kl} trials. Column l of group g has an indicator $G_{gl} \in \{0, 1, s\}$ with weights w: structural zero (every block of the group-column has $p = 0$), structural one ($p = 1$), or free; $w \sim \mathrm{Dir}(\kappa\pi)$, $\pi = (\pi_s/2, \pi_s/2, 1 - \pi_s)$, exchangeable over the relation's group-columns. In a free group-column of two or more rows each block has its own indicator H_{kl} with the fixed weights π (as in v0.7–v0.8); a one-row group-column is its block ($H = G$). A free block has $p_{kl} \sim \mathrm{Beta}(a_k, b_k)$. With $m_0 = \mathbf{1}[S = 0]$, $m_1 = \mathbf{1}[S = T]$, $m_s = B(a_k + S, b_k + T - S)/B(a_k, b_k)$ for one block and $\bar{m} = \sum_h \pi_h m_h$: (a) the blocked Gibbs sampler that draws, given w, each G_{gl} with*

$$P(G_{gl} = 0) \propto w_0 \prod_{k \in g} m_0(S_{kl}, T_{kl}), \quad P(G_{gl} = 1) \propto w_1 \prod_{k \in g} m_1, \quad P(G_{gl} = s) \propto w_s \prod_{k \in g} \bar{m}(S_{kl}, T_{kl}),$$

and then $w \sim \mathrm{Dir}(\kappa\pi + n(G))$*, leaves the posterior of* (w, G) *invariant, with the* H *integrated out; (b) an indicator without exposure (*$T = 0$ *throughout) has all marginals equal to 1, so it is independent of the data and integrates out exactly: running the chain on exposed group-columns only has the same stationary law for* w*, and mixes faster; (c) the draw of a dataset decides each group-column with the last* w *of the chain and, inside a free group-column, each block with* π *(*`grouped_zoib_draw`*);* $\kappa = 3$ *by default (*`structure.structural_pooling`*);* $\kappa = 0$ *restores the fixed weights* π *of v0.7–v0.8 everywhere.*

Proof. (a) Given w the likelihood of group-column (g, l) under $G = 0$ or $G = 1$ is the product of its blocks' spike likelihoods; under $G = s$ the blocks' indicators are independent with weights π, and summing them out gives $\prod_k \bar{m}$, which does not depend on w. Given the G, w has the Dirichlet–multinomial conjugate posterior. Both are full conditionals of the model with H summed out, so the Gibbs sampler leaves that posterior invariant. (b) With $T = 0$ every marginal is 1, so the indicator's conditional is w itself and its factor in the posterior of w sums to 1 over its values. (c) is a definition; given $G = s$, drawing H with π is its exact conditional. □

Why only the group level is learned. The group-columns are the units the draw decides on, and they are exchangeable within a relation; the blocks inside a free group-column are not exchangeable with them, because their indicators also absorb how the segments mix rows of different structure. An earlier version learned one w from both levels. On bi (real data) the many partial blocks inside free group-columns drove w_1 to 0.14 (0 in one relation), so complete blocks with one or two trials were drawn from the slab with probability up to 0.6, and 4–7% of the reported values were lost (cardinality KS 0.26–0.46 against a limit of 0.12). With the row level kept at π, the expected number of present pairs equals that of fixed weights (62,498 against 62,481 of 62,761 observed in bi's largest relation), and w_1 learned on grouped thin rows equals the share of complete group-columns (0.290 against 0.292, `t_zoib_pooling`). In a relation whose 112 observed blocks are all complete (bi-like, ungrouped), fixed weights drew a free block somewhere in 99.7% of datasets (5.56 free blocks per dataset); learned weights draw 0.00. Data drawn from the model itself recover its weights within 3 posterior standard deviations. In the reporting scenario a filed header form is complete in 99.8% of datasets and 0.01% of item values carry another version (C26, C27); `t_fixed_weights_option` shows that $\kappa = 0$ gives back v0.8's exact structure.

Proposition 88 (The slab is estimated from partial entries)**.** *The free component of Prop. 87 must put little mass where the spikes are for the weights to be identified (George and McCulloch 1993). The empirical-Bayes slab of row* k *(Kleinman's moment estimator) is therefore estimated from the entries with* $0 < S < T$ *only, the entries that only a free block explains; a row with fewer than three such entries takes the slab estimated from the partial entries of the whole relation (rows exchangeable a level up), and* $\mathrm{Beta}(a_0, b_0)$ *is kept only when the relation itself has fewer than three. Up to v0.8 complete entries (*$S = T$*) entered the estimate, so a row of complete entries had a slab concentrated near 1, indistinguishable from the structural one: with weights learned from the data, complete blocks moved to the slab and were drawn with* $p \approx 0.99$ *(a header form not complete in 2% of filings).*

Proof. The moment estimator is that of Prop. 87's slab restricted to entries whose likelihood under both spikes is zero, so the estimate is a function of the free blocks only. If the slab has mass ϵ within δ of 1, a complete block of T trials has slab marginal at least $\epsilon(1-\delta)^T$, and the posterior odds of one against free are $w_1/(w_s\, m_s)$, which tend to w_1/w_s as the slab concentrates at 1: the data then cannot move the weights between the two components. □

15.2 Counts, inclusion and participation

Proposition 89 (Pólya-urn marginal in closed form)**.** *For counts $t_1..t_n$ under* $\mathrm{DP}(a, F_0)$ *with* $F_0 = \mathrm{Poisson}(\lambda)$*, f_0 its pmf, let m_x be the multiplicity of value x and $b_x = a f_0(x)$. Then*

$$\log p(t) = \sum_x \Big[\log b_x + \log\Gamma(b_x + m_x) - \log\Gamma(b_x + 1)\Big] - \log\frac{\Gamma(a+n)}{\Gamma(a)}.$$

Proof. By the urn, draw i equals x with probability $(b_x + c)/(a + i)$, c the earlier copies of x. The numerators for the m_x copies of x multiply to $\prod_{c=0}^{m_x-1}(b_x + c) = b_x\Gamma(b_x + m_x)/\Gamma(b_x + 1)$ and the denominators to $\Gamma(a + n)/\Gamma(a)$. Computing $\log b_x$ directly avoids the underflow of $a f_0(x)$ far in the Poisson tail, which made v0.8 return $-\infty$ (and a spurious Bayes factor) for a single large count. □

Proposition 90 (Inclusion grid for the sparse Dirichlet)**.** *In Prop. 19's representation the auxiliary $u \in (0, 1)$ has density $\propto u^{A_o-1}(1-u)^{n-1}\prod_j(1 + \rho u^{\alpha_j})$. With $y = \log(-\log u)$ the density of y is*

$$g(y) \propto \exp\big(A_o\ell + (n-1)\log(1-e^{\ell}) + y\big)\prod_j(1 + \rho e^{\alpha_j\ell}), \qquad \ell = -e^y,$$

which decays like $\exp(-A_o e^y)$ *as $y \to \infty$ and like e^{ny} as $y \to -\infty$; a grid on $[-60, 40]$ holds all but a negligible mass for every $A_o \geq 10^{-12}$ and $n \geq 1$.*

Proof. $u = \exp(-e^y)$, $|du/dy| = u\,e^y$, which gives g. As $y \to \infty$, $\ell \to -\infty$ and the factor $u^{A_o} = e^{A_o\ell}$ dominates; as $y \to -\infty$, $1 - u = 1 - e^{\ell} \sim e^y$, so $(1-u)^{n-1}e^y \sim e^{ny}$. The v0.8 grid on $\operatorname{logit} u$ decays only like $e^{A_o y}$ at $-\infty$ and lost most of the mass for small A_o. □

Check `t_sparse_dirichlet` compares the sampler's expected number of included unobserved levels with the exact posterior $P(|J| = j) \propto \binom{m}{j}\rho^j\Gamma(A_o + j\alpha)/\Gamma(A_o + j\alpha + n)$ in three regimes.

Proposition 91 (Pooled participation prior)**.** *The shares s_{0t} of parents without children in the segments t of a link are exchangeable: $s_{0t} \sim \mathrm{Beta}(cm, c(1-m))$ with $m = (\sum Z + \frac{1}{2})/(\sum N + 1)$ and c by the Beta–binomial moment estimator in $[1, 10^4]$; the per-dataset draw is $s_{0t} \sim \mathrm{Beta}(cm + Z_t, c(1-m) + N_t - Z_t)$. When no source parent is empty, $E[s_{0t}] \leq (\frac{1}{2}/(\sum N + 1))\,c/(c + N_t)\cdot(1 + o(1))$, instead of $1/(N_t + 2)$ under the flat per-segment prior of v0.7–v0.8.*

Proof. Conjugacy gives the posterior. With $Z = 0$ and $\tau^2 \leq 0$ the estimator gives $c = 10^4$, and the posterior mean is $cm/(c + N_t)$. □

The flat prior gave every small segment about $1/(N_t + 2)$ of empty parents, which is the 3% of AdventureWorks customers without a sale reported since v0.5; the fake order warehouse now has every customer with an order in every dataset (`t_mandatory_participation`).

Proposition 92 (Covering mandatory parents)**.** *When every parent needs at least μ children, the parent segments of the child rows are drawn i.i.d. from ϕ_z and accepted when every segment t receives $N_t \geq \mu M_t$ rows (rejection, up to `rules.max_rejection_rounds` times): an accepted draw has exactly the conditional law given the constraint. If no draw is accepted, rows are moved from segments with a surplus, the row moving from s to t chosen with probability proportional to $\phi_{z,t}/\phi_{z,s}$, until every deficit is filled; this moves exactly the total deficit and changes nothing when the constraint already holds.*

Proof. Rejection sampling from a product law restricted to an event yields the restricted law. The repair stops when every $N_t \geq \mu M_t$; each move lowers one deficit by one and never takes a source below its own need, so the number of moves is $\sum_t(\mu M_t - N_t)^+$. □

15.3 Exact conditioning on rules

Proposition 93 (Rules are checked on the draws that are kept)**.** *Let a fact row be $X = (Y, D)$ with Y the modelled columns and D the independent per-row draws (stated distributions, soft-rule indicators and offsets, group-effect priorities). Rejection per row that redraws (Y, D) of invalid rows, keeps (Y, D) of valid ones and evaluates the rules on that same (Y, D) yields rows with law $Q(X \mid \text{valid})$. v0.8 drew D afresh on a copy for the check and again for the output, so the output's D was independent of the check: rows with* qty < 2 *under the rule* qty ≥ 2 *(415 of 1,978 in the probe) were output.*

Proof. Each row's accepted value is the first valid one in an i.i.d. sequence from Q, which has law $Q(\cdot \mid \text{valid})$. The proof needs the checked and the returned value to be the same draw. The engine stores D in hidden columns with a per-row mask, draws only rows whose mask is unset (new or redrawn rows), and drops the hidden columns at the end of the dataset; recomputing a frame (as the SCD step does) reuses them. □

Proposition 94 (Unique keys over a link and attributes)**.** *The attributes of an entity's rows are drawn before its links. A unique key made of links and modelled attributes is therefore a constraint on the links given the attributes, and the Metropolis–Hastings kernel of the link moves (independence proposals from each row's own factor, accepted iff the new state is valid) targets the product law restricted to it, as for keys over links only. Attribute redraws in the finishing step keep it as part of their rejection.*

Proof. Given the attributes, the key's attribute part is fixed, so the set of valid link assignments is a fixed set; the acceptance ratio of an independence proposal from Q_i restricted to a fixed set is 1 on valid states, as in the keys-over-links case. □

v0.8 kept keys made only of links (link moves) or only of attributes (finishing), and silently ignored mixed keys of entities (26 duplicates of (product, balance) in `t_unique_moves_fast`).

15.4 Copula sample size

Proposition 95 (Pooled copula sample)**.** *The posterior standard deviation of a latent correlation fitted on n_{fit} rows is $\approx (1 - \rho^2)/\sqrt{n_{\text{fit}}}$ (measured 0.0171, 0.009, 0.0047 for $n = 3k$, 12k, 48k). The pooled copula of a table of n rows and p columns is now fitted on* $\max(2\,\text{cap}, \lfloor C/p^2 \rfloor)$ *rows, $C =$ `copula.pooled_cells` (10^6), within the $O(np^2)$ cost of Prop. 80's sampler, so that the dataset-to-dataset spread of a weak correlation is not several times its sampling error.*

Proof. The rank likelihood's Fisher information for ρ is $n/(1 - \rho^2)^2$ to first order (Hoff 2007), and each sweep costs $O(np^2)$, so C/p^2 rows fix the cost per sweep. □

Contoso's join association $z = 3.57$ (Quantity against order date) was a single dataset's posterior draw of $+0.021$ for a correlation whose posterior mean is -0.022, fitted on 3,000 of 224,000 rows.

15.5 Identification

Proposition 96 (Permutation test of a forest)**.** *Let $T(\pi) = 1$ when the column's pointers permuted by π (nulls included) form a forest. Under the null that the column's values are exchangeable over the rows, $p = (1 + \#\{b \leq B : T(\pi_b) = 1\})/(B + 1)$ with $\pi_1..\pi_B$ uniform permutations satisfies $P(p \leq \alpha) \leq \alpha$ for every α. The engine uses $B = 199$, $\alpha = 0.01$ (smallest $p = 0.005$), stops early once $p > \alpha$ is certain, and accepts the self-reference when $p \leq \alpha$; otherwise it asks when no row points to itself.*

Proof. Under exchangeability the observed column is one more draw from the permutation distribution, so its rank among the $B + 1$ values is uniform (ties broken in its favour), which gives the stated p-value its validity (Phipson and Smyth 2010). Stopping when $1 + \# > \alpha(B + 1)$ only skips draws that cannot change the decision. □

A code column with values 1..3 in a 2,000-row table was accepted as a hierarchy in 124 of 200 samples under Prop. 78's uniform null; with the permutation test, 0 of 60.

Proposition 97 (Inclusion of strings by chance)**.** *Let the $v \geq 3$ distinct child strings in K use the length set $\mathcal{L}$ and character classes Σ (capitals, small letters, digits, each counted if used, plus every other character seen), $A = |\Sigma|$, $w = \sum_{L \in \mathcal{L}} A^L$ and $k = |K \cap \Sigma^{\mathcal{L}}|$. Under the null that the v values are a uniform draw without replacement from the code space, $P = \binom{k}{v}/\binom{w}{v} \leq (k/w)^v$ ($P = 1$ when $k \geq w$; $P = 1$ for $v \leq 2$). Inclusion is evidence when $P \leq 10^{-6}$.*

Proof. Hypergeometric, as Prop. 76, with the bound from $(k-i)/(w-i) \leq k/w$. v0.8 returned $P = 0$ for any three strings, so three section codes $\{A, B, C\}$ inside grades $A..F$ were a link; now $\log_{10} P = -2.11$. □

Names need values. For a link between two tables, names count as evidence only with randomness $R \geq R_{\text{crit}}(v) = \min(0.5, 1 - 2/\sqrt{v})$ over the v included values, or $P \leq 10^{-6}$; the same R_{crit} replaces the fixed 0.6 of the question rule. $R_{\text{crit}}(v)$ lies below the simulated 0.1% quantile of R for random samples of $v = 5, 10, 20, 50$ values. Composite keys and self-references are now confirmed on the whole table, not on a sample.

15.6 Calibrated limits of the scorecard

Proposition 98 (Posterior-predictive null of a distribution difference)**.** *Let the synthetic sample of size m be drawn from a Bayesian bootstrap of the real sample of size n (the posterior predictive of a saturated model). For each x, $E[F_{\text{syn}}(x) \mid \textit{real}] = F_n(x)$ and*

$$\operatorname{Var}\big(F_{\text{syn}}(x) - F_n(x) \mid \textit{real}\big) = F_n(x)\big(1 - F_n(x)\big)\Big(\frac{1}{n+1} + \frac{1}{m}\Big).$$

The two-sample effective size $1/(2/n + 1/m)$ of v0.8 therefore overstated the null spread by $\sqrt{(2/n + 1/m)/(1/(n+1) + 1/m)}$ (≈ 1.22 at $n = m$); KS, TVD, null-share, association and join limits now use $1/(n+1) + 1/m$ with the effective (Kish) sizes.

Proof. With Dirichlet$(1, \ldots, 1)$ weights W, $G(x) = \sum_i W_i \mathbf{1}[x_i \leq x]$ has mean $F_n(x)$ and variance $F_n(1-F_n)/(n+1)$ (the variance of a sum of Dir(1^n) components over a subset of size nF_n). Given G, $F_{\text{syn}}(x)$ is a binomial share with variance $G(1-G)/m$, whose expectation is $F_n(1-F_n)(1-\frac{1}{n+1})/m$; to first order the total is as stated. □

Proposition 99 (Maximum over a family)**.** *For K centred Gaussian statistics $Z_1..Z_K$ with unit variances and any correlations, $P(\max_k |Z_k| \leq z) \geq \prod_k P(|Z_k| \leq z)$ (Šidák 1967). The limit $z_K = \Phi^{-1}(1 - \frac{1}{2}(1 - (1-\alpha)^{1/K}))$ therefore keeps the family's false-failure rate at most α; for $K = 48$ and $\alpha = 0.01$, $z_K = 3.71$ (v0.8 used 3 for the maximum over up to 48 cells).*

Proof. Šidák's inequality, and $\prod_k P(|Z_k| \leq z_K) = (1 - (1 - (1-\alpha)^{1/K}))^K = 1 - \alpha$. □

The same correction is used for the worst association pair and across tables for the discriminator, whose null standard deviation is bounded by that of one fold's Mann–Whitney AUC, $\sqrt{(n+1)/(3n^2)}$ (within a fold the scores come from a model fitted on the other fold, so the Mann–Whitney null law is exact; the mean of two correlated folds has at most that variance). Novel combinations use the Good–Turing missing mass N_1/n with Esty's variance (Good 1953; Esty 1983), and unequal expected cardinalities compare binomially thinned counts (thinning a Poisson count keeps it Poisson, Rényi 1956). The scorecard reports the expected number of false failures, $\sum \alpha_i$, so a run of 1,303 checks is read against about $0.01 \times 1303 \approx 13$ expected chance failures.

16 Fitting inside the database (v1.0)

Version 1.0 packages the engine for production use and fits fact tables larger than memory from statistics computed where the data lives (*pushdown*). A fact table is *pushed down* when it is a fact that no table references, has no type 2 history, and is not the source of an aggregate column that the source does not store (`relsynth/pushdown/planner.py`). Its rows never leave the database as a whole: the engine reads (i) exact aggregates computed by `GROUP BY` and window functions, and (ii) a uniform sample of at most `synthesis.pushdown.sample_rows` rows. Every other table is loaded as before. The propositions below show that this changes nothing in the relational model, and that the attribute models fitted on the sample are an exact posterior given the sampled rows. The checks are in `tests/unit/test_sqlfuncs_stats.py`, `tests/unit/test_pushdown.py` and `tests/integration/test_servers.py`; they run on SQLite, PostgreSQL 16, SQL Server 2022 and Oracle Database 26ai Free (23.26).

16.1 What the relational model needs from a fact

Let a pushed fact F have N rows, key columns $k_1..k_m$ (links to dimensions or to a parent fact, the parts of a composite key) and continuous columns $c_1..c_p$. For a non-null value x_i of column c with average rank r_i among the N_c non-null values let $z_i = \Phi^{-1}(r_i/(N_c+1))$ (its *normal score*), and $z_i = 0$ for a null. For a key tuple u write

$$n(u) = \#\{i : k(i) = u\}, \qquad S_c(u) = \sum_{i:k(i)=u} z_i^{(c)}, \qquad Q_c(u) = \sum_{i:k(i)=u} (z_i^{(c)})^2,$$

with a null key value forming a group of its own.

Proposition 100 (Sufficiency of the in-database statistics)**.** *Given the loaded tables (entities and non-pushed facts), the full conditional of every latent quantity of the relational model of Sections 3–15 that involves a pushed fact F—segment assignments of the entities it links, block and link rates (Props. 5, 6), count and multiplicity laws (Props. 21, 32, 89), participation and inclusion (Props. 34, 91, 19), nested refinement (Prop. 27), order headers (Prop. 43) and running balances (Prop. 49)—depends on the rows of F only through*

1. *the counts $n(u)$ over the key tuples the model uses (pairs of links, a composite key with its links, a header key with its header columns);*
2. *the score sums $S_c(u)$ and $Q_c(u)$ for the (at most three) continuous columns that segment discovery uses;*
3. *for an order header, the first row of every document key in primary-key order and the number of cells that differ from it; for a running balance, the first row of every group in the rule's order, the number of non-null values and the minimum;*
4. *N, the number of non-null values of each key, and the distinct referenced values of each key.*

Proof. Each listed model block is a product over the rows of F of a factor that depends on a row only through its key tuple u and, for segment discovery, through the rank features $z^{(c)}$ of the row. Block-rate and multiplicity likelihoods are products of Poisson or multinomial terms whose exponents are the counts of rows per pair of linked rows, so they are functions of $n(u)$ (the conjugate posteriors of Props. 5 and 21 use sums of $n(u)$ over segment pairs). The rank-feature factor of segment discovery is Gaussian in the scores, with a mean per pair of segments and a common variance per column (`PairRel`, `OccRel`), so its log-likelihood over the rows of a key tuple is $a\,n(u) + b\,S_c(u) + c\,Q_c(u)$ for coefficients that do not depend on the rows, and the conjugate updates of the means and variances use the same three sums. Participation and inclusion use which keys occur (the distinct values) and how often ($n(u)$). A nested fact is linked to its parent

fact through its composite key, so the counts over (composite key, links) give the parent row and the dimension rows of every group, with the multiplicity $n(u)$ as the group's weight. An order header is the first row of its document key (Prop. 43); its split is lossless exactly when no cell differs from that row, which is what (3) counts. A running balance needs its opening value per group (the first row in the rule's order) and its floor (the minimum); the rest of the column is a deterministic function of the flows (Prop. 49). Nothing else of F enters these factors. □

Proposition 101 (Agreement with in-memory fitting)**.** *Let the loaded rows of every table be put in a canonical order: by primary key, else by the first declared unique key that holds, and by a hash of the row's values when there is neither (`canonical_order`); and let order headers split from a fact be put in the order of their document key. If the sample of a pushed fact holds all its rows, the fitted model and every synthetic dataset are identical, value for value, to those of fitting the whole table in memory with the same seed; and the model fitted from a database does not depend on the order in which the database returns rows.*

Proof. By Prop. 100 the relational blocks see the same numbers, computed either by `FrameStats` (pandas) or by `SqlStats` (SQL); the attribute models see the same rows. The fit is a deterministic function of its inputs, the seed and the order in which rows are visited (the Gibbs sweeps visit rows in order). The canonical order (by primary key, else by the first declared unique key, else by a hash of the row; order headers split from a fact by their document key) is a function of the set of rows only, so two backends, or two scans of one table, give the same input sequence. The only remaining difference is floating-point: SQL sums are accumulated in another order than in pandas, and servers compute Φ^{-1} with Prop. 102; both perturb S_c, Q_c by relative errors below 10^{-8}, far under the resolution at which a segment draw changes. The checks compare the synthetic tables cell by cell: identical for all four pushed fact shapes on SQLite (`test_full_sample_pushdown_equals_in_memory`) and for a pushed star schema on PostgreSQL, SQL Server and Oracle against SQLite (`test_full_sample_pushdown_matches_sqlite_in_memory`). □

Without the canonical order the same database gave different models on different servers (SQL Server and Oracle return rows in a different order than SQLite and PostgreSQL), and a server using parallel scans could give different models on two runs. v1.0 orders every loaded table; this changes in-memory results relative to v0.9 by a permutation of the input rows, which leaves the posterior unchanged and changes individual draws only.

16.2 Normal scores on database servers

Proposition 102 (Acklam's quantile approximation)**.** *PostgreSQL, SQL Server and Oracle have no standard-normal quantile function. relsynth computes $\Phi^{-1}(p)$ in SQL with Acklam's rational approximation (a rational function of $q = p - \frac{1}{2}$ on $[0.02425, 0.97575]$ and of $\sqrt{-2\log p}$ in the tails), whose relative error is below $\delta = 1.15 \times 10^{-9}$ on $(0, 1)$. Hence $|\hat{S}_c(u) - S_c(u)| \le \delta \sum_{i:k(i)=u} |z_i|$ and $|\hat{Q}_c(u) - Q_c(u)| \le (2\delta + \delta^2)\, Q_c(u)$. SQLite registers the exact function (`scipy.special.ndtri`) as a user function.*

Proof. The error bound of the approximation is Acklam's (2003), and `test_acklam_error_bound` confirms it in double precision on 4,081 points spanning $[10^{-12}, 1 - 10^{-12}]$ (relative error where $|\Phi^{-1}(p)| > 10^{-3}$, absolute error below 10^{-8} everywhere). For a sum, $|\sum \hat{z}_i - \sum z_i| \le \sum |\hat{z}_i - z_i| \le \delta \sum |z_i|$ with $\delta = 1.15 \times 10^{-9}$; for squares $|\hat{z}^2 - z^2| = |z|^2\,|(1+\epsilon)^2 - 1| \le (2\delta + \delta^2) z^2$. □

16.3 The sample

Proposition 103 (Seeded hash sampling is a Bernoulli sample)**.** *The sample keeps row i when $h(s, \kappa_i) < \lfloor q\, 2^{32} \rfloor$, where κ_i is the row's primary key (all its columns when there is none), s the seed, $q = \min(1, n/N)$ for the requested size n, and h the database's hash reduced to*

$[0, 2^{32})$ *(`hashtextextended` on PostgreSQL, SHA-256 by `HASHBYTES` on SQL Server, `ORA_HASH` on Oracle, BLAKE2b on SQLite and files). If $h(s, \cdot)$ behaves as a uniform random function independent of the data, then (a) the inclusion indicators are i.i.d. Bernoulli(q) and independent of every column value, so the sample is a uniform random subset with $|S| \sim \mathrm{Bin}(N, q)$; (b) the same seed gives the same sample on every run and every engine with the same hash; (c) $P(||S| - Nq| > 8\sqrt{Nq(1-q)} + 10) < 1.3 \times 10^{-15}$, so the guard that stops a sample of that size (a broken hash, such as one returning NULL) does not stop a valid sample in practice.*

Proof. (a) Under the assumption, $h(s, \kappa_i)$ are i.i.d. uniform on $[0, 2^{32})$ because the keys are distinct, so each comparison is Bernoulli with probability $\lfloor q2^{32} \rfloor / 2^{32}$, which differs from q by less than 2^{-32}. (b) The comparison is a deterministic function of (s, κ_i). (c) With $\sigma^2 = Nq(1-q)$ and $t = 8\sigma + 10$, Bernstein's inequality gives $P(||S| - Nq| > t) \le 2\exp(-t^2/(2\sigma^2 + 2t/3)) \to 2e^{-32} < 2.6 \times 10^{-14}$ as $\sigma \to \infty$; the exact binomial tails, computed for $N \in \{10, \ldots, 10^9\}$ and $q \in [10^{-6}, 1 - 10^{-6}]$, are at most 1.24×10^{-15}, the normal value $2\Phi(-8)$ approached for large σ (for small σ the additive 10 makes the tail far smaller). `test_python_hash_is_uniform_and_seeded` tests uniformity, and `test_sample_is_uniform_reproducible_and_sized` size and reproducibility on each engine. □

Proposition 104 (Attribute models from the sample are an exact posterior)**.** *Let the sample be drawn as in Prop. 103. The posterior of the attribute-model parameters θ of the pushed fact given the sampled rows,*

$$p(\theta \mid x_S, S) \propto p(\theta) \prod_{i \in S} p(x_i \mid \theta, keys_i),$$

is the exact posterior given the observed data: the sampling mechanism is ignorable. Synthetic datasets are therefore posterior-predictive draws (Prop. 66) given the sampled rows for the attributes and given all rows for the relational part; their sizes follow the exact N, not $|S|$. The posterior is wider than with all rows (its variance scales with $1/|S|$ for regular parameters), so the scorecard's real side uses exact statistics where they exist (row counts, null keys, cardinalities, duplicate rates, pair multiplicities) and the sample elsewhere, with limits computed for the sample's size.

Proof. The inclusion indicators are independent of the values and of θ (Prop. 103a), so $p(S \mid x, \theta) = p(S)$ and the likelihood of the observed data factors as $p(S) \prod_{i \in S} p(x_i \mid \theta)$; the factor $p(S)$ does not involve θ (Rubin 1976: data missing completely at random). Prop. 100 makes the relational blocks depend on all rows through exact statistics. The sizes use N because the scale rule (Prop. 7) is applied to the exact counts. □

16.4 Why entities are always loaded

Proposition 105 (Eligibility)**.** *A table cannot be fitted from aggregates alone when any of the following holds: it is an entity (each of its rows carries its own latent segment, whose full conditional needs that row's attributes and links, Prop. 47); another table references it (it is then an entity of its own rows for that table's links, including the parent of a nested fact); it has type 2 history (Prop. 72 draws versions per business key and row); or an aggregate column of another table is computed from its rows and the source does not store that column (the column must be recomputed from the rows). Every other fact is eligible, including nested facts, facts split into order headers, bases of views and facts with running balances.*

Proof. For each excluded case the named proposition's full conditional has a factor per row of the table that is not a function of the statistics of Prop. 100. For the eligible shapes, Prop. 100 lists the statistics that replace the rows. □

Memory and time. A pushed fact costs memory proportional to the number of distinct key tuples plus the sample, instead of rows times columns: on AdventureWorks (all ten eligible facts pushed, including the 776,286-row inventory fact with its running balance) the peak resident memory fell from 2.1 GiB to 1.6 GiB at a sample of 30,000 rows, and the run took 668 s instead of 803 s on

SQLite. The database does the grouping; the number of round trips is a constant per statistic and does not grow with N.

16.5 Sizes of child entities

An entity C whose rows link to a parent entity P (orders to customers) has its own size factor s_C; the link model draws each of its rows' parents from the segments of P. If only P is resized, the expected number of C rows per P row changes by s_C/s_P, which is rarely what a user who asks for "ten times the customers" wants. From v1.0 an entity without a factor of its own takes the factor of the first parent it links to that has one (`inherit_scale`, applied in topological order, so a chain of entities follows its root), and the scorecard expects the same sizes. With $s_C = s_P$ the expected rows per parent are those of the source, and facts anchored on C follow by Prop. 7. A table given its own factor keeps it. *Test* `test_child_entities_follow_a_scaled_parent`: customers $\times 3$ gives exactly $3\times$ the invoices and $3\times$ the lines within 10%, with every invoice-per-customer check passing.

17 Evaluation thresholds

A synthetic sample of size m drawn from a model fitted on n real rows deviates from the real sample by the real sample's own noise plus the fit's noise plus the synthetic sampling noise, so the effective size for two-sample tests is $n_{\text{eff}} = (2/n + 1/m)^{-1}$. The scorecard uses: the two-sample KS critical value $\sqrt{-\frac{1}{2}\log(\alpha/2)}\, n_{\text{eff}}^{-1/2}$ at $\alpha = 1\%$ (at least 0.08); for categorical columns 2.5 times the expected TVD, $\frac{1}{2}\sum_l \sqrt{2p_l(1-p_l)/(\pi n_{\text{eff}})}$ (at least 0.08); for association differences $2.5\, n_{\text{eff}}^{-1/2}$ (at least 0.15); for joins, the size of the ancestor table rather than the number of joined rows, since a join repeats each ancestor row. Without this, a table of 12 departments fails on noise alone, as the first runs showed. Since v0.3, rows whose level occurs in fewer than `suppress_small_counts` rows are left out of association pairs (a single record's coincidence is not a dependence the model should copy, and copying it would be a disclosure), the correlation ratio between a categorical and a numeric column is computed on ranks, and rows per partner are compared after rescaling by the realised size of the grand anchor (tree and grain tables have random sizes). Grain metrics report the duplicate rate of each unique key, the share of repeated pairs, and the share of synthetic rows whose combination of two reference keys never occurs in the real data where the real data covers its combinations densely. Since v0.4 every limit is at least the sampling limit of the quantity it tests: null-rate differences use $2.5\sqrt{p(1-p)/n_{\text{eff}}}$; an association pair uses the rows that carry information on it (non-null rows, and for a categorical the rows away from its mode: a flag set on 5 rows has 5 such rows, however long the table); joined columns use Kish's effective size $(\sum w)^2/\sum w^2$ of the ancestor rows weighted by how many fact rows each carries; tree fan-out and attribute-by-depth use the KS and TVD limits at the tree's size. Check C41 measures the false-alarm rate of the null-rate and association limits on independent samples from one law (0 of 400 at $n = 59$).

18 Checks without ground truth (real data)

On real data the segments and parameters are unknown, so recovery checks are replaced by checks that need none. `local/replicate_checks.py` runs them where the data lives and writes only aggregates:

R1 the scorecard of Section 17 (marginals, dependence, discriminator AUC, joins, cardinalities, trees, time, text, privacy, rules);

R2 copula convergence: split $\hat{R}$ of every correlation and of $\log\det C$, and the largest change of a correlation between chain halves (and the Metropolis acceptance in `bayes` mode);

R3 posterior predictive checks (Proposition 30, a parameter redraw before each copy): for rows per parent, share of parents without rows, dispersion, tree depth and leaves, busiest-month share and fact means, the real value must lie within 3 standard deviations of D synthetic copies (or within 3%);

R4 the scale rule (Proposition 7) on the user's own schema;

R5 structural validity: FK integrity and acyclic hierarchies;

R6 stability: agreement (ARI) of segmentations from two seeds, the real-data stand-in for recovery;

R7 reproducibility.

Privacy. For entity tables with at least 3 modelled columns and 50 rows: exact-copy rate (at most max(1%, 2× the real table's own duplicate rate)) and the distance-to-closest-record ratio, the 5th percentile of synthetic-to-real distances over the 5th percentile of real-to-real leave-one-out distances, in rank-normal and one-hot space (at least 0.5; on coarse tables split as in Proposition 31). Direct identifiers never pass through the model (`pii` columns are regenerated from formats). *Check C04*: no copies, ratio 0.98, no shared emails.

Rehearsal. `local/make_sqlite_demo.py` loads the enterprise scenario into SQLite with declared keys; introspection then drafts the spec from the catalog alone (Section 19).

19 Diagnostics loop

The bundle holds the spec (credentials masked), a column profile without values or level names, the model summary, the scorecard, the replication checks and any error traceback. `relsynth doctor` applies rules D01–D22 and R2–R6, each mapping a failed metric plus profile evidence to a cause and a spec patch (for example: a categorical column that is 95% e-mail addresses → `pii`; all segments occupied and checks on that table failing → double the cap; rare text copied → raise k, shorten L; rows per parent too dispersed and a small parent table → frailty or copy mode). Version 0.3 adds: rows per parent *less* dispersed than Poisson → a grain, with the unique key proposed from the profile's key candidates (D03, D21); a broken unique key (D18); repeated pairs (D19); combinations of reference keys the real data never has (D20, stratify); a stored computed column that disagrees with its rule (D22); and slow copula mixing → rank mode, longer chains, then a higher rare-column threshold (D07). Reference tables in copy mode are exempt from the leak rules. Issues whose cause is in the code (failed steps, broken foreign keys) are marked *engine* and come back to the project with their traceback. `relsynth tune` loops run → doctor → apply. *Check C19*: four planted spec errors in the enterprise schema (email declared categorical, segment cap 1, text guards off, hierarchy depth capped at 3) are all identified from the bundle alone; tuning removes all four and raises the scorecard from 117/129 to 122/122 within three rounds.

20 Changes to layer 1 found while building and testing

1. **Measures conditional on parents.** Segment-level means (layer 1's m_{kl}) lose the link between a sale amount and the product's own price; parent attributes are now conditioning columns of the fact copula (Section 4).

2. **Degree correction.** Pure Poisson blocks under-disperse rows per parent; Gamma frailties, rescaled within segments (Proposition 6(c)), fix this without breaking the scale rule. Estimating a frailty needs ≥ 30 rows in the segment: an estimate from 6 executives once killed most synthetic hierarchies.

3. **Segment marginals** use a Dirichlet-process posterior instead of falling back to the table marginal below a row threshold, and a weak prior ($a = 0.25$): with $a = 2$, 7% of synthetic rows in a 6-row segment took the wrong level.

4. **Small tables** draw segment labels by largest-remainder allocation of $n^*\pi$ rather than multinomially, and reference tables can be copied.

5. **Strongly dependent categoricals** are modelled jointly (Remark 2); nominal levels are ordered by their most associated column.

6. **Columns constant within a segment** are excluded from that segment's copula, whose correlations they cannot inform.

7. **Trees** use posterior predictive checks and a depth limit from the data.

8. **Text guards** count rows, not occurrences, so common phrases are allowed and rare details are not.

9. **(v0.3) Grain.** A unique key of two foreign keys makes a table an occupancy model of its parents' product, not a Poisson count (Propositions 22–24).

10. **(v0.3) Structural zeros** in every prior that gives a combination positive mass: occupancy, activity, null rates, category levels and extra keys (Propositions 17–19), pooled within strata.

11. **(v0.3) Computed columns** are computed, never sampled, on both sides (Proposition 16).

12. **(v0.3) Nullable keys** are a level, not a reason to drop the row (Proposition 26).

13. **(v0.3) Copula**: the rank likelihood with parameter expansion replaces Metropolis on vine partial correlations as the default, with chain extension by $\hat{R}$, and rare columns are kept out (Propositions 28, 29).

14. **(v0.4) Rows per parent** are chosen between Poisson/NB and a DP by Bayes factor, for facts and for links between entities, and DP counts of entity links are drawn given the segment totals (Propositions 32, 33); parents that always have children use 1+Poisson (Proposition 34).

15. **(v0.4) Categorical hierarchies** are joint categoricals over their observed combinations (Proposition 36).

16. **(v0.4) Tiny tables** are stratified by an own column and generated with their stratum counts; hierarchies are conditioned on their size (Propositions 37, 38).

17. **(v0.4) Values**: few-valued numbers are categoricals on their values, tail levels never leak a placeholder, aggregates keep the stored precision (Proposition 40).

21 Limitations and open items

- **Copula mixing** is solved by the parameter-expanded rank sampler (Proposition 28); `bayes` mode (vine Metropolis) remains available and still mixes slowly for near-deterministic pairs.
- **Structure the data cannot show.** A rule that holds in every real row but is not declared (e.g. "a filed form always has its header items") is reproduced statistically, not exactly (S8: 99.8%). Declaring it as a check rule makes it exact (Proposition 14).
- **Within-segment dependence is Gaussian.** Non-monotone relations inside a segment are captured only through more segments. ClavaDDPM and TabDDPM remain the baselines to compare against on real data (layer 1 item 9).

- **Text is n-gram based**: fluent for templated enterprise text (tickets, notes), weak for long free prose. The foundation-model route needs a model endpoint inside the data perimeter.
- **The prototype is in memory.** Layer 1's in-database statistics (GROUP BY pushdown, sketches) are the layer 3 path to 10^8-row facts; every quantity the model uses is already an aggregate of that form.
- **Formal DP** (layer 1 Algorithm 6) is not wired into this prototype; the privacy checks here are empirical.
- **Four real databases so far** (the reporting model, Chinook, AdventureWorks DW, Contoso V2). Their generic lessons are built in and tested on S8–S11 and C57–C68; other real schemas will test other situations.
- **Unknown members on facts** are restored under a missing-at-random assumption (Proposition 73); when unknown references depend on other columns the member should be modelled as an ordinary row.
- **The frailty shape** is an empirical-Bayes (moment) estimate, not a posterior draw; the participation hurdle (Proposition 70) corrects the zero share it can miss.
- **Stated column distributions** are drawn independently of the other columns.
- **Hierarchies of a handful of rows** (8 employees) are learned from one example of each relation. Even with strata and size conditioning, about one draw in five places a manager under the wrong person; the reproducible fix is to declare the hierarchy as reference data if it is not confidential.
- **Regularities the source generator imposed** (each track bought at most twice, cells spread evenly over submissions) are under-dispersion on the partner side; the engine models counts per anchor, not per partner, and reports these.

22 Results

All 79 of 79 checks pass (507 of 507 individual assertions). Each row names the property it tests; the per-assertion values are in `checks/results.md`.

Table 1: Layer 2 checks on fake data with known ground truth.

Id	Check	Property	Items	Pass
C00	Layer 1 algorithm checks still pass	Layer 1 paper, Table 4	1/1	yes
C01	Copula synthesis preserves every marginal (mixed types, nulls)	Prop. 1, 2	8/8	yes
C02	Posterior of the latent correlation covers the true copula	Prop. 3 (Alg. 2 of layer 1)	4/4	yes
C03	Missingness that depends on other columns is reproduced	Prop. 2 (null as lowest latent bin)	4/4	yes
C04	Single-table privacy: no copies, distance to closest record	Sec. 9	3/3	yes
C05	Star schema: latent segments of both dimensions are recovered	Prop. 4 (identifiability up to relabelling)	2/2	yes

Id	Check	Property	Items	Pass
C06	Star schema: block rates eta_kl inside their posterior intervals	Prop. 5	2/2	yes
C07	Rows per customer follow the fitted Poisson mixture (Poisson splitting)	Prop. 6	3/3	yes
C08	Scale rule: resizing the partner keeps rows per anchor; resizing the anchor scales the fact	Prop. 7 (layer 1 scale rule)	4/4	yes
C09	Snowflake: segments recovered at every level and carried down the chain	Prop. 8	7/7	yes
C10	Self-referencing tree: multi-type Galton-Watson fit and finiteness; level-conditioned generator	Props. 9, 10, 54	5/5	yes
C11	Self-referencing tree: synthetic hierarchy is valid and matches the source shape	Prop. 10, Sec. 6	5/5	yes
C12	Temporal fact: segment-specific seasonality recovered and reproduced	Prop. 11	4/4	yes
C13	Text: proper predictive distribution, no rare content leaks, class signal kept	Prop. 12, 13	6/6	yes
C14	Enterprise schema end to end: integrity, rules, full scorecard	Sec. 7, Prop. 14	4/4	yes
C15	Rejection sampling for business rules gives the exact conditional law	Prop. 14	2/2	yes
C16	Gamma frailty gives the negative-binomial variance mu + mu^2/alpha	Prop. 6 (degree correction)	2/2	yes
C17	Dirichlet-process marginal posterior: mean is (a F0 + n Fn)/(a + n)	Prop. 2	1/1	yes
C18	Same seed gives an identical synthetic database	Sec. 7 (reproducibility)	1/1	yes
C19	Doctor: finds planted spec errors from the bundle alone and the fixes improve the result	Sec. 8 (diagnostics loop)	6/6	yes
C20	Rule language: three-valued logic, SQL null semantics and string functions	Prop. 15 (three-valued rules)	9/9	yes
C21	Structural (spike-and-slab) priors: samplers match the exact posterior	Props. 17-19 (spike-and-slab priors)	13/13	yes
C22	Empirical-Bayes slabs: Beta-binomial concentration recovered, shrinkage lowers error	Props. 20-21 (empirical-Bayes slabs)	4/4	yes
C23	Dropping duplicate rows of a unique key is exact conditioning on presence	Prop. 24 (unique keys)	2/2	yes
C24	Pair model selection: occupancy for grain data, Poisson for count data	Prop. 22 (repeat test)	4/4	yes
C25	Rank copula: the fit depends on the order of the data only	Prop. 28 (rank likelihood)	3/3	yes
C26	Reporting scenario end to end: grain, keys, aggregates, rules, scorecard	Props. 16, 23, 25, 26 combined (S8); Prop. 52 (structural zeros)	8/8	yes
C27	Reporting scenario: omitted and partial filings reproduced	Props. 18, 27 (grouped zeros, nested refinement)	3/3	yes
C28	Reporting scenario: open rows repeat, closed items never do	Props. 21, 24 (multiplicity, unique keys)	3/3	yes

Id	Check	Property	Items	Pass
C29	Grain entity scales with its anchor and stays unique	Prop. 23 (grain entities)	4/4	yes
C30	Introspect drafts keys, links and grains from PascalCase warehouse names	Sec. 12 (introspect)	6/6	yes
C31	Doctor proposes the missing grain from the bundle alone	Sec. 12 (doctor D03, D21)	2/2	yes
C32	Computed columns are filled on the source and stored ones are audited	Prop. 16 (computed columns)	4/4	yes
C33	Posterior predictive draws: redraw gives spread that covers the source	Prop. 30 (posterior predictive)	3/3	yes
C34	Rare columns are kept out of the copula and the chain converges	Prop. 29 (rare columns)	3/3	yes
C35	Count-model choice by Bayes factor: DP only for counts no Poisson/NB mixture explains	Prop. 32 (count-model evidence)	3/3	yes
C36	DP counts given a total: the pairwise Gibbs sampler draws the exact conditional	Prop. 33 (conditioned counts)	3/3	yes
C37	Functional dependencies: city -> state -> country kept exactly, no extra disclosure	Prop. 36 (dependency groups)	5/5	yes
C38	Tiny parent: strata by its own column keep who receives children; hierarchy keeps its size	Props. 37, 38 (attribute strata, conditioned tree)	4/4	yes
C39	Small shop end to end: invoices per customer, lines per invoice, lookups, aggregates, precision	Props. 32-35, 39, 40 with Prop. 16 (S9)	10/10	yes
C40	Introspect on a CSV shop: lookups, aggregate with a product, anchors, grains, party tables, questions	Sec. 12 (introspect)	8/8	yes
C41	Evaluator limits hold their nominal level on small and clustered samples; no placeholder leaks	Sec. 10 (evaluation), Prop. 40 (tail levels)	5/5	yes
C42	Doctor proposes a grain only when independence would have produced repeats	Prop. 41 (grain evidence)	5/5	yes
C43	Sales warehouse end to end: one header per order, lines 1..n, date gaps, identities, stock, who buys what	Props. 43-45, 48, 49, 53 (headers, offsets, refill, tied segments, running balances, mandatory lines)	18/18	yes
C44	Introspect on the sales warehouse (SQLite, declared keys): calendar, headers, offsets, lookups, identities, running balance, first purchase; same spec twice	Prop. 42 (significance of discovered rules)	10/10	yes
C45	Same seed gives the same data in a new process; per-row shares keep count dispersion	Prop. 46 (posterior-draw shares of copied rows)	3/3	yes
C46	Warehouse catalog regressions: binary columns, composite keys, partly declared links, order header, validation, money TVD, pooled level after reordering	Findings F01-F08, F22-F24 of the AdventureWorks log	10/10	yes
C47	Introspect drafts reference tables, keyed and mapped lookups, views, line numbers from 0, date gaps, implied dimensions, ratio columns, cross-table rules and key order	Props. 55-58; findings K01-K16 of the Contoso run	16/16	yes

Id	Check	Property	Items	Pass
C48	Synthesis keeps views, keyed and mapped lookups, implied dimensions, ratio factors, date gaps, line numbers, key order and cross-table rules exactly, without biasing the marginals	Props. 55-59	26/26	yes
C49	Building blocks: date functions, keyed lookups across date storage, ratio factors at the coarsest precision, computed-column cycles, file formats	Props. 55-58; findings K03, K09, K14, K15	8/8	yes
C50	Introspect on Parquet files without declared keys and with renamed columns: line keys, the view with its renamed columns, the dimension code not asked as a foreign key; the view rebuilt exactly	Props. 55, 60; findings K02, K04, K15	7/7	yes
C51	Links under a rule across two parents: exact draws per row keep the rule on every row, the source's cell margin, and the source's law of each parent within its cell	Props. 59, rake; finding K10	4/4	yes
C52	Conditional inclusion on a fact's anchor: facts that only reference some parent rows (quotas only for sales staff) never reference the excluded rows, and keep their row counts	Prop. anchorincl; AW finding (quotas linked to non-sales staff)	15/15	yes
C53	Labels nested in a coarser code (cities in states): drafted as whole-value text within the code; synthetic cities sit in their source states, with no label held by fewer than 3 source rows	Prop. nested; finding K17	4/4	yes
C54	DP row counts under mandatory participation: the base measure 1 + Poisson keeps every parent with a child, with the same mean, in unconditional and total-conditioned draws	Prop. mandatory DP; finding K18	5/5	yes
C55	A small dimension with no label to anchor on anchors itself: the code stays modelled (keeps its dependence on the other columns) and the columns it determines are read by its value	Prop. dimension (self-anchored); finding R6	3/3	yes
C56	Conditional inclusion on an occupancy fact's partner: order lines never reference products that are never sold (null price), even when one segment mixes them with sold products, and keep lines per order	Prop. anchorincl (partner side); AW regression (one reseller line on a product without a price)	6/6	yes
C57	One posterior draw per dataset: a dataset is reproducible from its index; across datasets the spread of a share and of rows per parent is posterior plus sampling variance (about twice the sampling variance), and about the sampling variance alone in posterior-mean mode	Prop. postpred (v0.7)	6/6	yes
C58	A rule that holds on most rows (an age on a reference date) is found by the interval sweep and synthesised as a mixture: the share that keeps the rule and the offsets of the rest match the source	Prop. soft (v0.7)	5/5	yes
C59	A value shared by all rows of a group (a daily price factor per product category) is found by introspect and drawn once per group: every synthetic group has one factor, and the factor law matches	Prop. groupfx (v0.7)	4/4	yes

Id	Check	Property	Items	Pass
C60	Two links that point at parents in the same area more often than chance (customer and store) are found by introspect and synthesised as a diagonal-inflated mixture: the same-area share matches and the country rule still holds	Prop. affinity (v0.7)	5/5	yes
C61	Participation of parents in an entity's link is a hurdle: the share of customers without an order matches the source (half of them, and none when every customer has one)	Prop. hurdle (v0.7)	9/9	yes
C62	A latent segment never mixes parent rows that may have children with rows that never do (only sales staff have quotas): the eligibility column stratifies the segments, so the children per eligible parent match the source in every dataset, even with one segment allowed	Prop. strata (v0.7)	4/4	yes
C63	A type 2 slowly changing dimension (versions of a product code with valid-from/to dates and a current flag) is found by introspect and synthesised per natural key: version counts match, stable attributes repeat across versions, each version ends the day before the next starts and only the latest is open	Prop. scd2 (v0.7)	10/10	yes
C64	An unknown member (key -1, 'Unknown') that facts point at is found by introspect, kept out of the model, copied, and referenced with the source's share	Prop. unknown (v0.7)	3/3	yes
C65	A bridge with allocation weights (accounts of a customer, weights summing to 1) is found by introspect; synthetic weights sum to 1 in every group and keep their spread	Prop. bridge (v0.7)	3/3	yes
C66	Recipes of the specification: copy every dimension and synthesise only the fact; draw a column from a stated distribution; name the anchor of a role-playing fact by its key column	spec-reference.md (v0.7)	7/7	yes
C67	The specification is strict: a misspelt key anywhere is an error with the closest valid key, and every spec drafted or written for the earlier tests still validates	schema.py (v0.7)	2/2	yes
C68	The annotated example of the specification reference (every v0.7 option in one spec) validates, runs, and keeps each option's property	spec-reference.md (v0.7)	8/8	yes
C69	Optimised kernels compute exactly what they replace	Prop. 'Exact kernels': same IEEE operations in the same order and the same random stream	8/8	yes

Id	Check	Property	Items	Pass
C70	Keys and links identified from the data, declared keys hidden	Props. 'Inclusion by chance' and 'Randomness bound': no false link; every missed link is asked	6/6	yes
C71	Identification statistics: chance inclusion, randomness, forest test	Props. 'Inclusion by chance', 'Randomness bound', 'Chance of a forest'	11/11	yes
C72	Exported model: verified, strict JSON, draws are the engine's datasets	Props. 'Exported draw m is the parameter of dataset m' and 'Verification conditions'	12/12	yes
C73	Standalone replica: keys, sizes and fidelity	Prop. 'The replica samples the core generative process'	10/10	yes
C74	Posterior blocks agree with Stan	Props. 'Block-rate posterior', 'Spike-and-slab Bernoulli', 'Rank likelihood with parameter expansion'	1/1	yes
C75	v0.9 engine: pooled weights, exact conditioning, coverage, warnings	Section 'v0.9' propositions	14/14	yes
C76	v0.9 scorecard calibration and fidelity statistics	Props. 'ppnull', 'sidak'	14/14	yes
C77	v0.9 identification: permutation forest test, chance inclusion	Props. 'forestperm', 'stringchance'	14/14	yes
C78	v0.9 input, output and security hardening	Spec reference	28/28	yes

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